



NTNU – Trondheim
Norwegian University of
Science and Technology

TET4185: Power Markets, Resources and Environment

Grid access

Congestion Management

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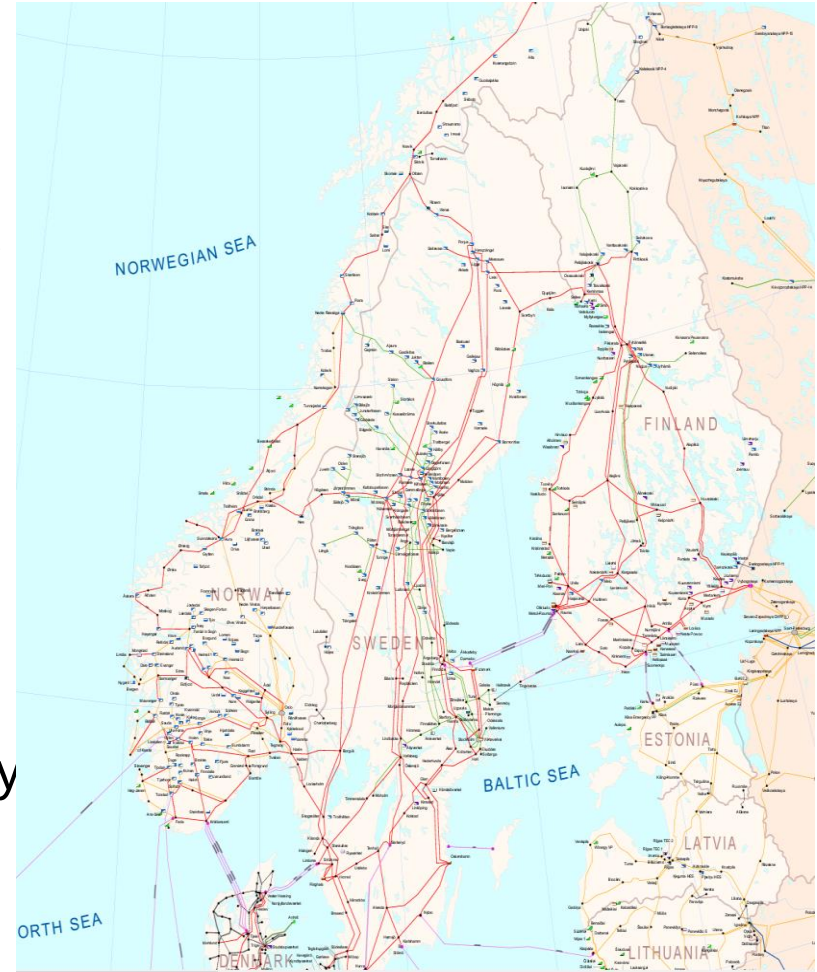


Perspective

- Congestion management
- Transfer capacity
- Congestion management methods
 - Area pricing
 - Counter trading
- Area pricing as an optimization Problem

Transfer Capacity Limitations

- The grid integrates geographically dispersed markets
- There is a maximum amount of energy that may be exchanged from one location to the next
- When this limit is reached, one talks about **congestion** and prices for connected areas will differ
- Transfer capacity limitations are directly related to network constraints and operational practice



Congestion Management

Management of constraints (bottlenecks) in the grid where there is insufficient transfer capacity to exchange energy.

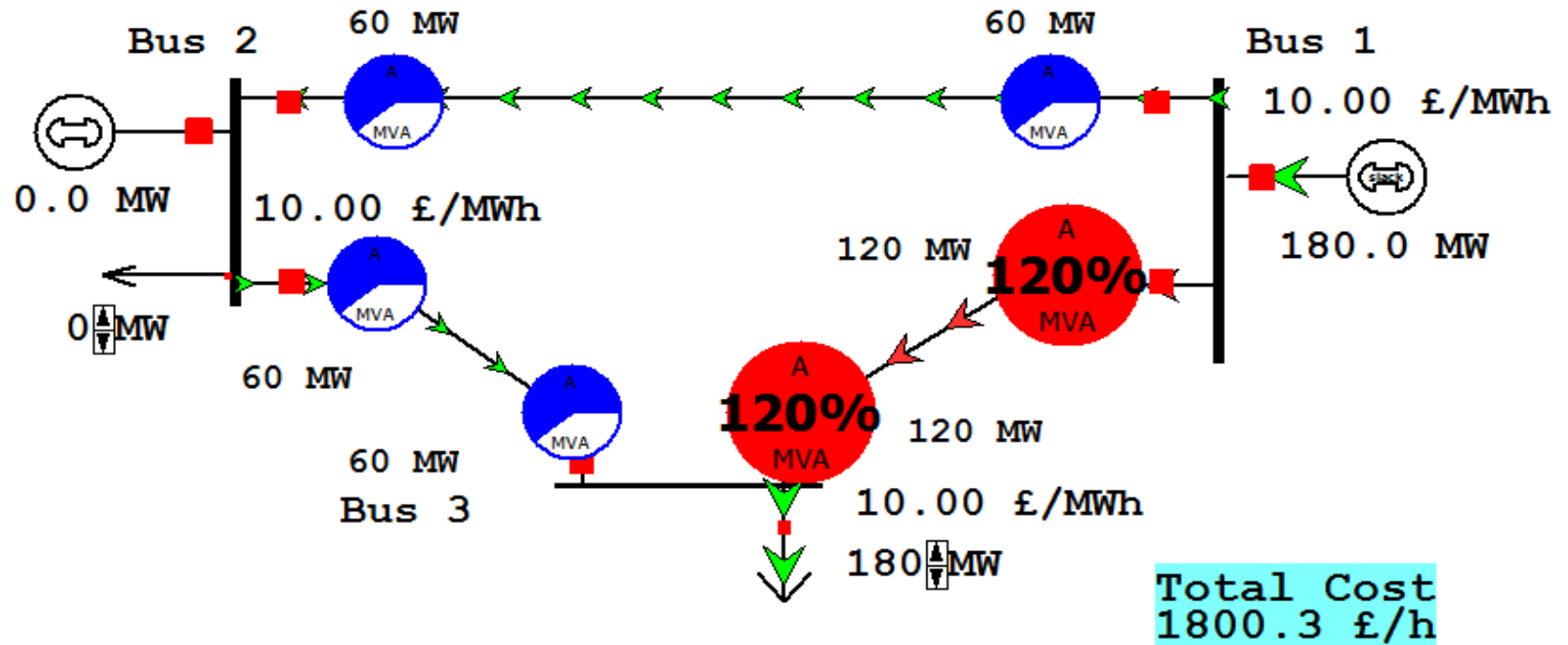
- Very important issue in market design because:
 - Crucial for system security
 - inadequate handling of transmission constraints may lead to overload → cascading overload → system collapse / blackout
 - Economic efficiency of the power system
 - wrong / too low transfer capacity leads to inefficient system dispatch (use of expensive vs. cheap generation)
 - Potential impact on the number of different prices in the system
 - one price, area / zonal, nodal
 - Market participants ability to trade across the system
 - Market power issues
 - ... and a lot more

Congestion Example

Maximum transmission capacity for every line is 100 MW

Gen 2's cost
is £12 per MWh

Gen 1's cost
is £10 per MWh



Line from Bus 1 to Bus 3 is over-loaded; all buses have same marginal cost

What is transfer capacity?

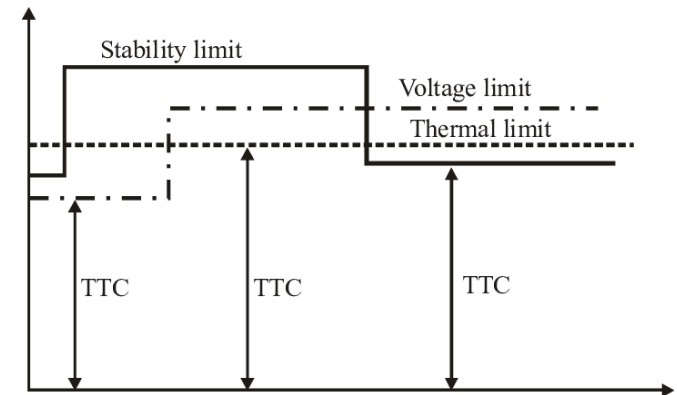
- ENTSO-E : European Network of Transmission System Operators for Electricity
 - NTC (Net Transfer Capacity)
 - $NTC = TTC - TRM$, where:
 - TTC: Total Transfer Capacity (TTC) → The maximum exchange programme between two adjacent areas compatible with operational security standards
 - TRM: Transmission Reliability Margin
 - ATC: Available Transfer Capacity
 - $ATC = NTC - AAC$, where:
 - AAC: Already Allocated Capacity

https://eepublicdownloads.entsoe.eu/clean-documents/pre2015/ntc/entsoe_proceduresCapacityAssessments.pdf

What determines Total Transfer Capacities?

Security of Supply: N-1 criterion → we can lose one component and full supply can still be maintained

- Dynamic constraints, system stability
- Voltage stability
- Thermal limit



Important: transfer capacity is not a simple, fixed number, but depends on very many factors

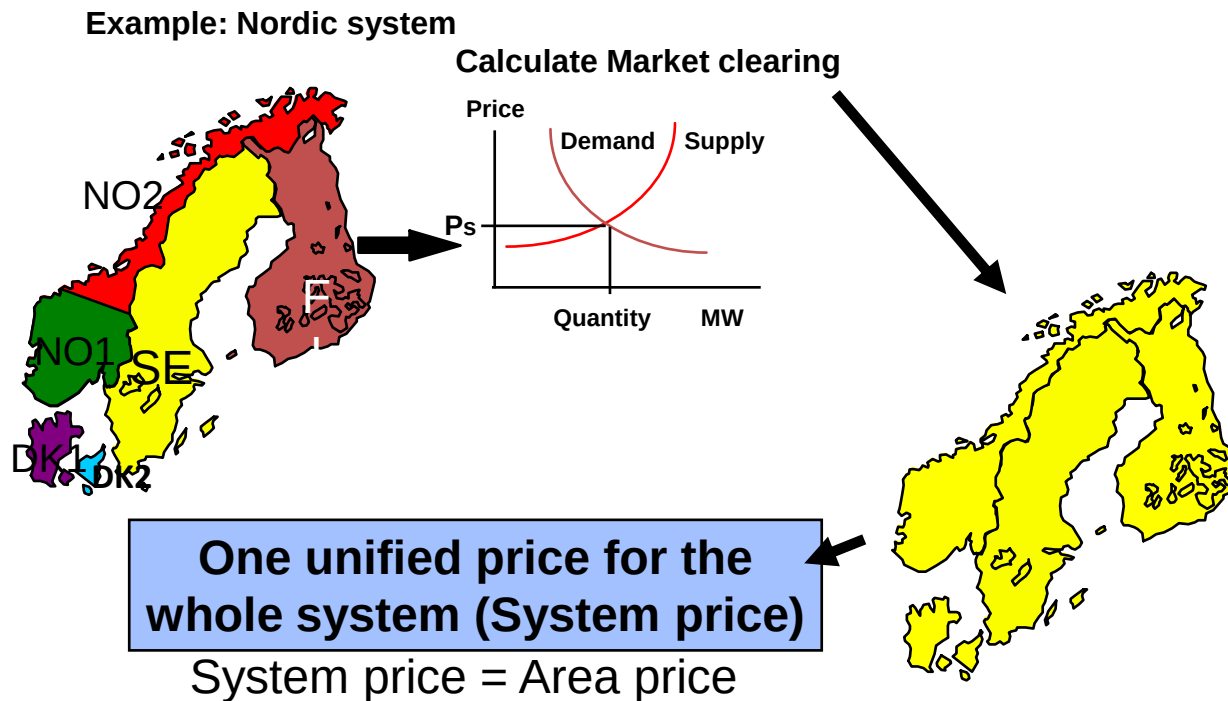
- comprehensive analyses required

https://eepublicdownloads.entsoe.eu/clean-documents/pre2015/ntc/entsoe_proceduresCapacityAssessments.pdf

Congestion Management Methods

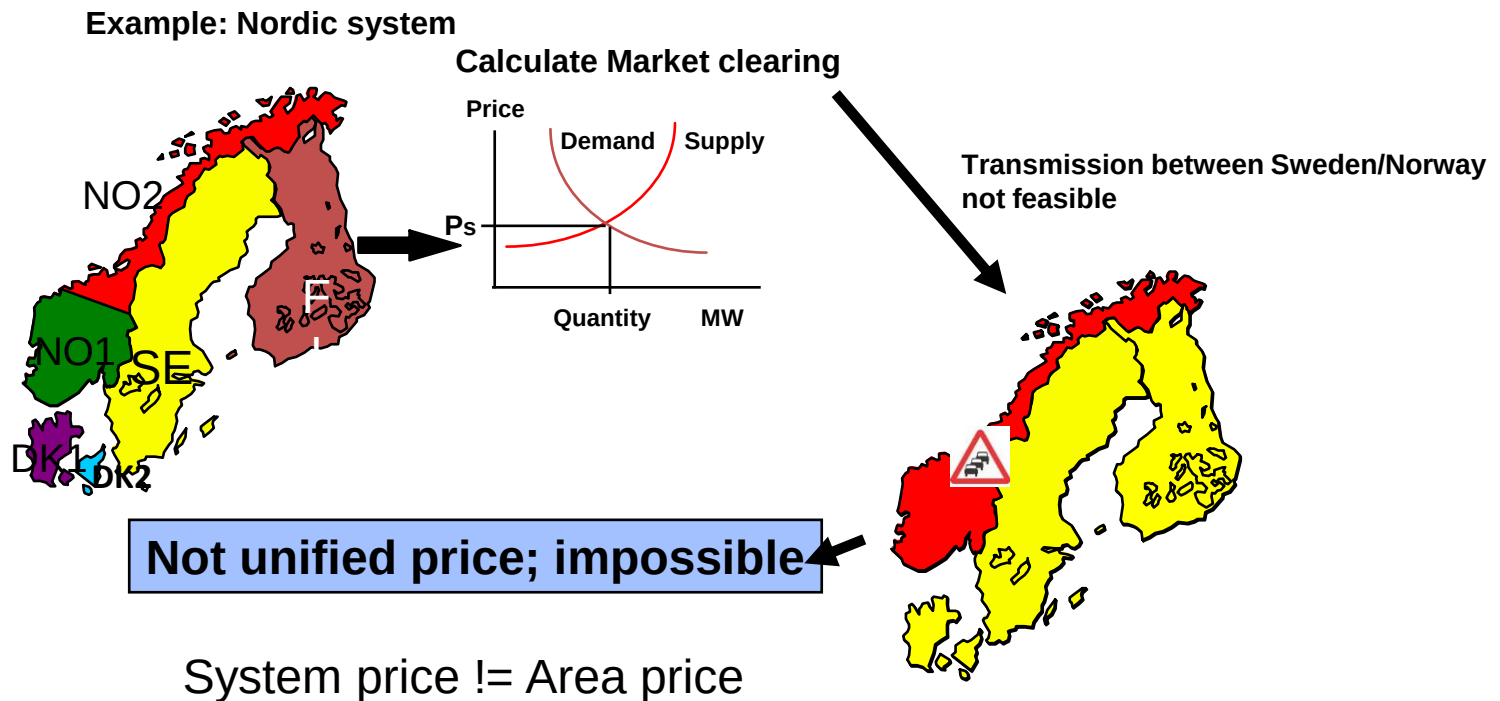
- **Area (or zonal) pricing , Market splitting**
 - grid is split in a limited number of areas, each consisting of many nodes
 - different prices in each zone
- **Counter trading , buy-back**
 - one price
 - congestion handled through transactions between TSO and involved generators (or consumers)
- **Nodal pricing**
 - based on Optimal Power Flow (OPF) calculation
 - different prices in each node

System price – Electricity price without physical limits



Source: Nord Pool

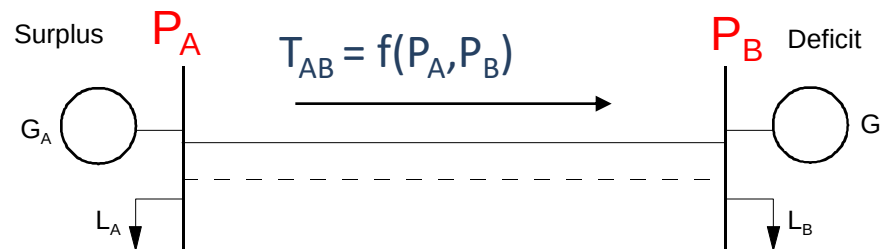
System price – Electricity price without physical limits



Source: Nord Pool

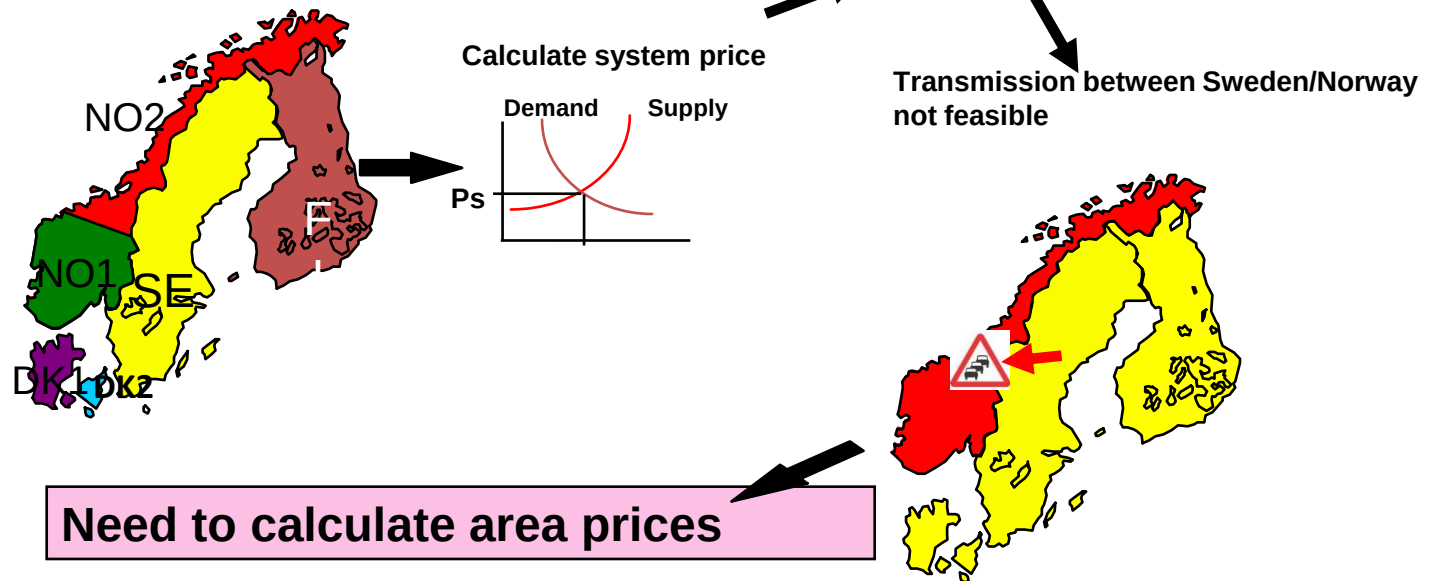
Area pricing

- If there is sufficient transfer capacity the power will flow from the surplus to deficit area without significant price difference
- Optimal congestion management: The prices are adjusted so that the exchange power matches the available transfer capacity

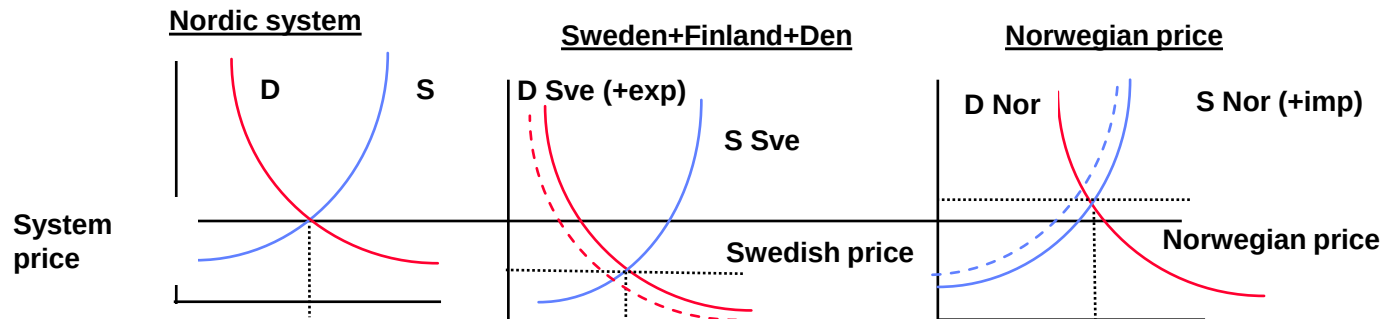


Area pricing

Example: Nordic system



Source: Nord Pool



System price $\neq$ Area price

Nordpool – System versus Area prices



Market Data

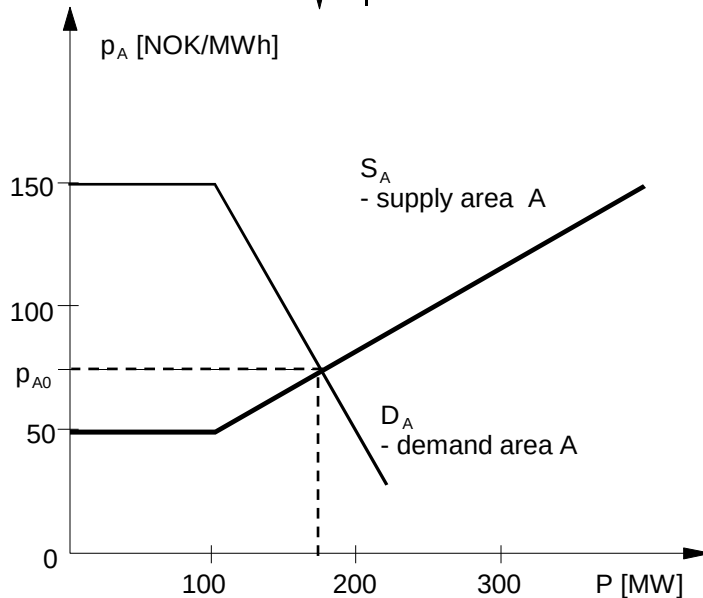
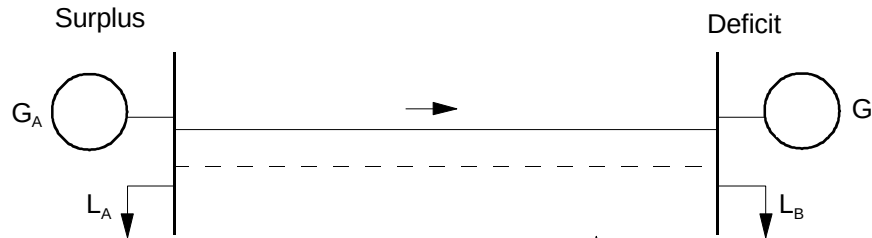
SYS € 48.79				
Delivery date 16 February 2024				
SE1 € 32.81	SE2 € 32.81	SE3 € 32.81	SE4 € 36.81	NO1 € 55.30
Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024
NO2 € 55.30	NO3 € 42.58	NO4 € 32.81	NO5 € 55.30	DK1 € 55.03
Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024
DK2 € 41.10	FI € 32.43	EE € 55.80	LT € 55.80	LV € 55.80
Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024
PL € 75.24	AT € 61.97	BE € 61.60	FR € 60.75	GER € 61.06
Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024	Delivery date 16 February 2024
NL € 64.04	UK £ 61.28			
Delivery date 16 February 2024	Delivery date 16 February 2024			

Source: Nord Pool

What we will cover now

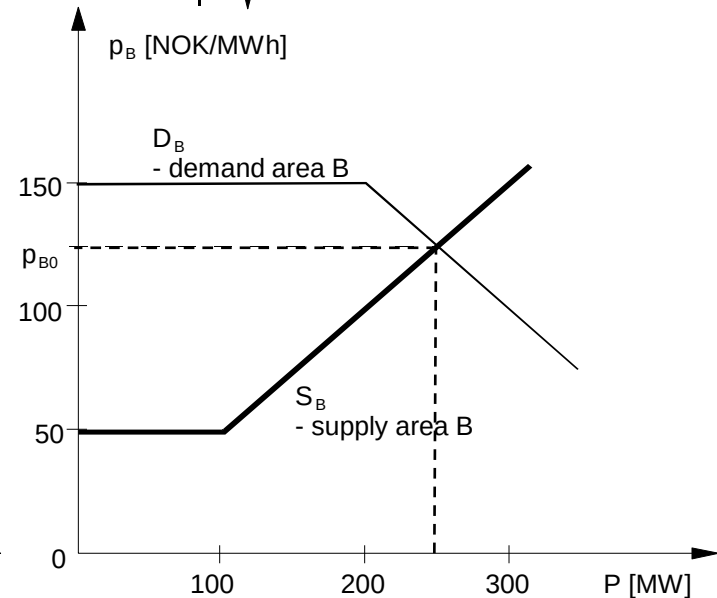
- How are area prices found?
- How does transmission limitation influence area prices?
- How do zonal pricing influence the price on each other?
- What happens with the money from the transferred capacity of different price zones?

Area pricing (example): simple two bidding zones



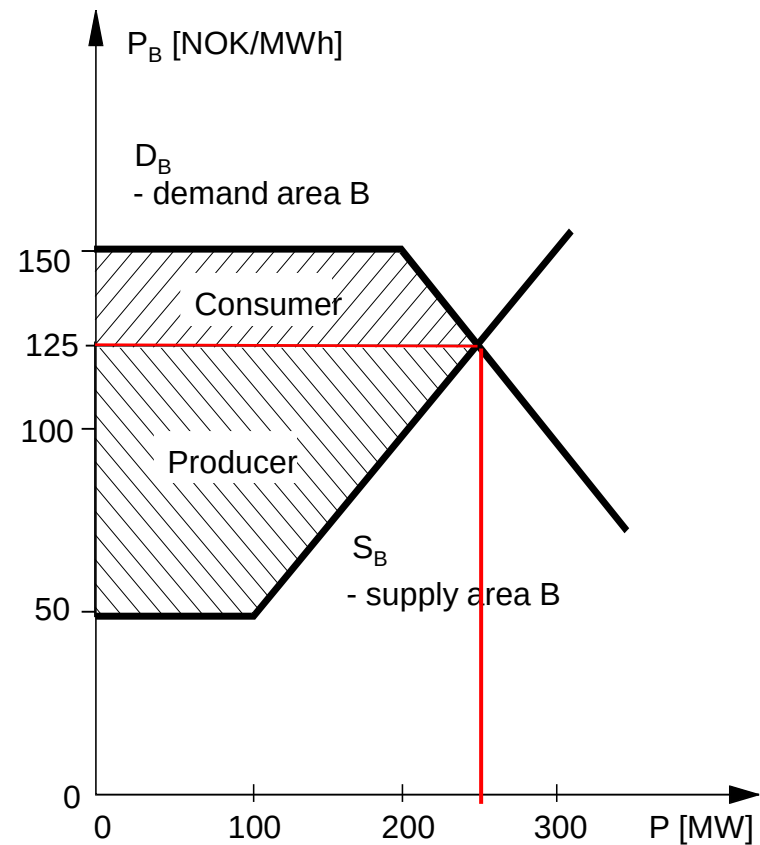
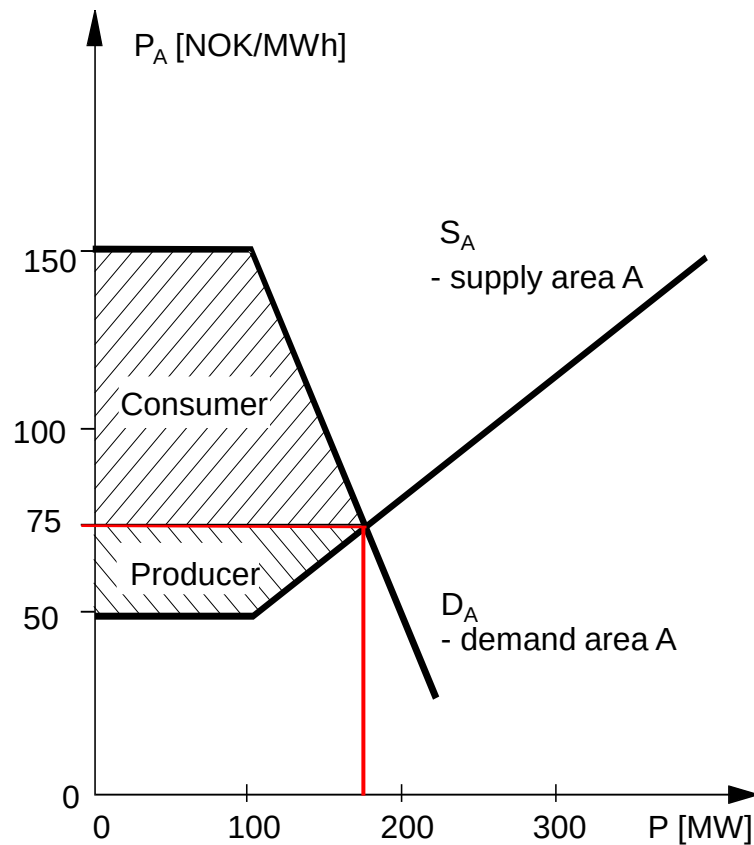
$$\text{Demand in A: } \begin{cases} p_A = 150 & \text{for } D_A \leq 100 \\ p_A = 250 - D_A & \text{for } D_A \geq 100 \end{cases}$$

$$\text{Supply in A: } \begin{cases} p_A = 50 & \text{for } S_A \leq 100 \\ p_A = 50 \frac{1}{3} + \frac{1}{3} S_A & \text{for } S_A \geq 100 \end{cases}$$



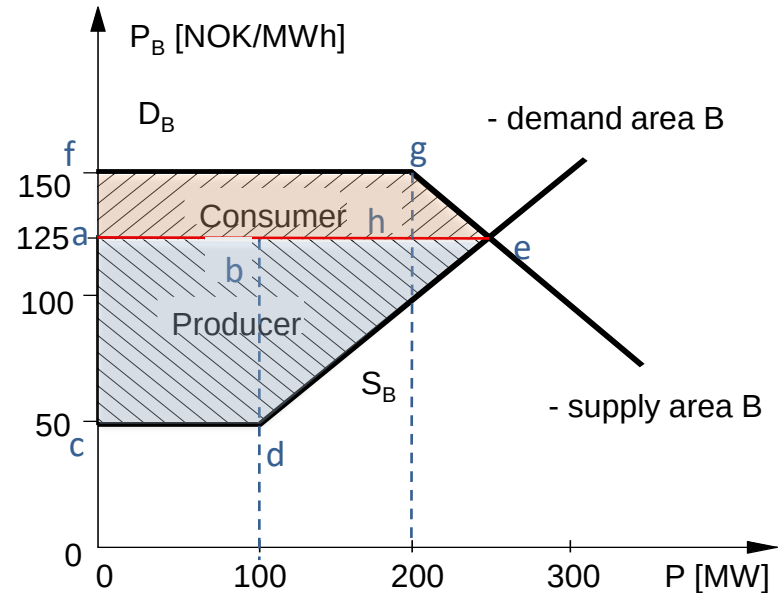
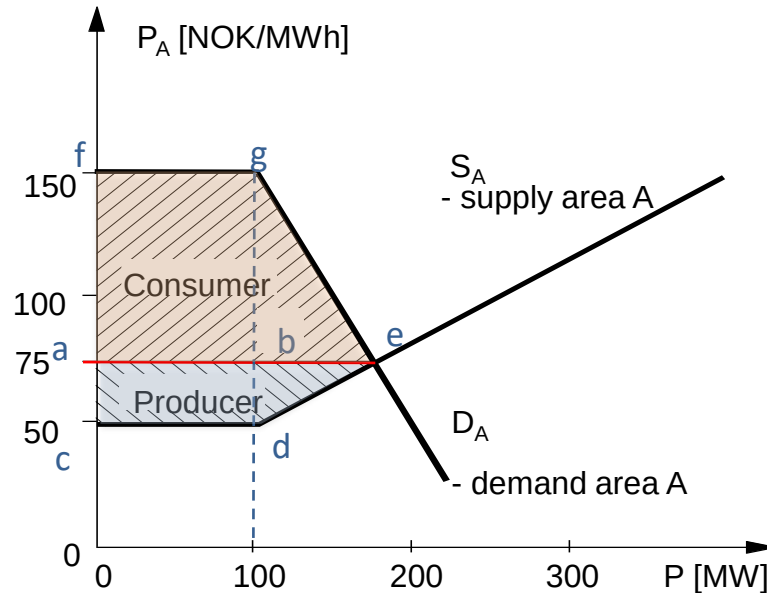
$$\text{Demand in B: } \begin{cases} p_B = 150 & \text{for } D_B \leq 200 \\ p_B = 250 - 0.5 D_B & \text{for } D_B \geq 200 \end{cases}$$

$$\text{Supply in B: } \begin{cases} p_B = 50 & \text{for } S_B \leq 100 \\ p_B = 0.5 S_B & \text{for } S_B \geq 100 \end{cases}$$



- Market equilibrium in bidding zone **A**: price $p_A=75$ NOK/MWh, $S_A = D_A = 175$ MW
- Market equilibrium in bidding zone **B**: price $p_B=125$ NOK/MWh, $S_B = D_A = 250$ MW

Consumer and Producer surplus



$$PS_A = (75 - 50) \cdot \left(100 + \frac{1}{2} \cdot 75 \right) = 3437.5 \text{ NOK}$$

$$PS_B = (125 - 50) \cdot \left(100 + \frac{1}{2} \cdot 150 \right) = 13125 \text{ NOK}$$

$$CS_A = (150 - 75) \cdot \left(100 + \frac{1}{2} \cdot 75 \right) = 10312.5 \text{ NOK}$$

$$CS_B = (150 - 125) \cdot \left(200 + \frac{1}{2} \cdot 50 \right) = 5625 \text{ NOK}$$

Total surplus: 32500 NOK

Merging the areas into a unified system



$$\text{Demand : } \begin{cases} p_A = 150 & \text{for } D_A \leq 100 \\ p_A = 250 - D_A & \text{for } D_A \geq 100 \end{cases}$$

$$\text{Demand : } \begin{cases} p_B = 150 & \text{for } D_B \leq 200 \\ p_B = 250 - 0.5D_B & \text{for } D_B \geq 200 \end{cases}$$

$$\text{Supply : } \begin{cases} p_A = 50 & \text{for } S_A \leq 100 \\ p_A = 50 \frac{1}{3} + \frac{1}{3} S_A & \text{for } S_A \geq 100 \end{cases}$$

$$\text{Supply : } \begin{cases} p_B = 50 & \text{for } S_B \leq 100 \\ p_B = 0.5S_B & \text{for } S_B \geq 100 \end{cases}$$

$$\text{Demand } \begin{cases} \text{Demand A: } p = 150 \text{ for } D \leq 100 \text{ \& } D = 250 - p \text{ for } D \geq 100 \\ \text{Demand B: } p = 150 \text{ for } D \leq 200 \text{ \& } D = 500 - 2p \text{ for } D \geq 200 \end{cases}$$

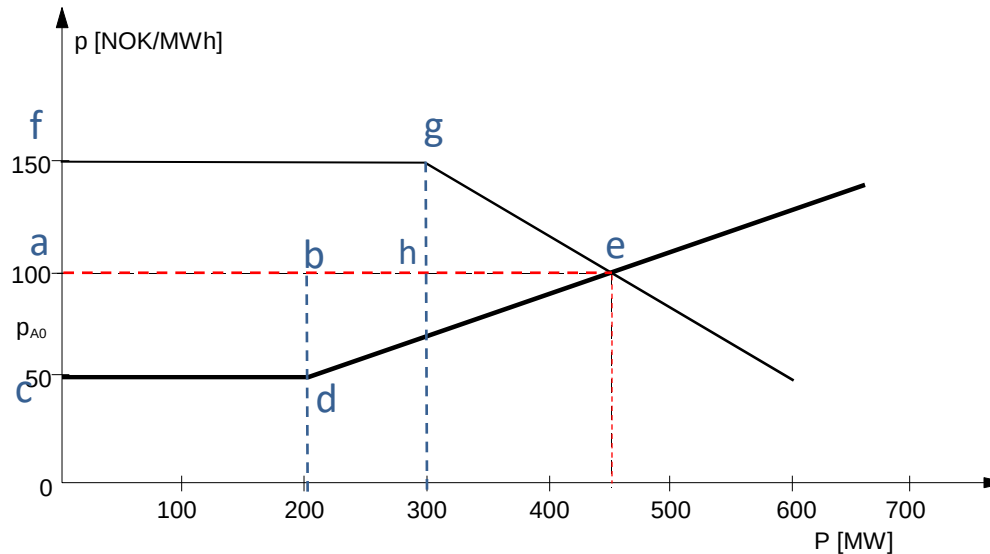
$$\text{Demand : } p = 150 \text{ for } D \leq 300 \text{ \& } D = 750 - 3p, \quad p = 250 - \frac{1}{3}D \text{ for } D \geq 300$$

$$\text{Supply } \begin{cases} \text{Supply A: } p = 50 \text{ for } S \leq 100 \text{ \& } S = 3p - 50 \text{ for } S \geq 100 \\ \text{Supply B: } p = 50 \text{ for } S \leq 100 \text{ \& } S = 2p \text{ for } S \geq 100 \end{cases}$$

$$\text{Supply: } p = 50 \text{ for } S \leq 200 \text{ \& } S = 5p - 50, \quad p = 10 + \frac{1}{5}S \text{ for } S \geq 200$$

No Transmission Constraints

Add demand and supply curves horizontally



$$\text{Demand : } \begin{cases} p = 150 & \text{for } D \leq 300 \\ p = 250 - \frac{1}{3}D & \text{for } D \geq 300 \end{cases}$$

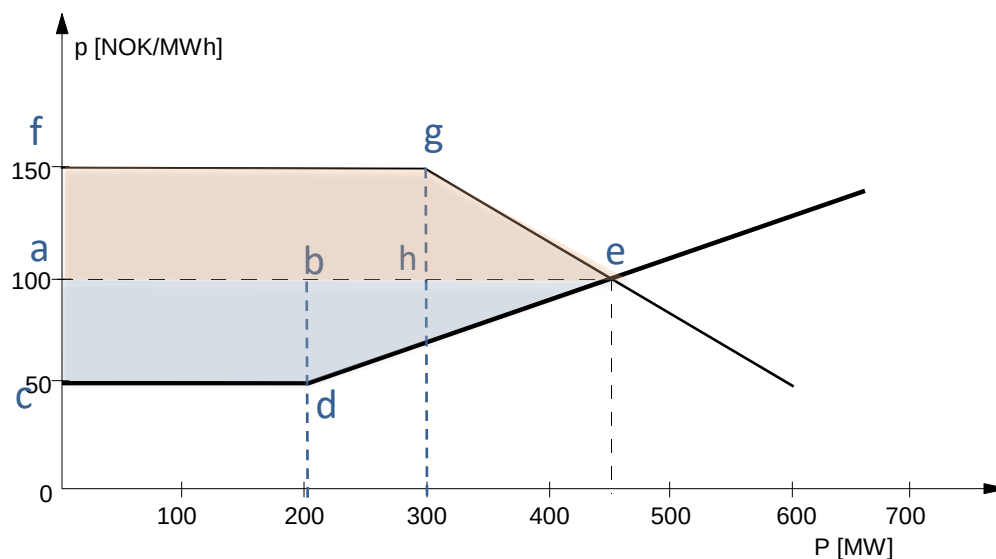
$$p = 100 \text{ NOK/MWh}$$

$$S=D= \underline{450} \text{ MW}$$

$$\text{Supply : } \begin{cases} p = 50 & \text{for } S \leq 200 \\ p = 10 + \frac{1}{5}S & \text{for } S \geq 200 \end{cases}$$

**More Capacity is accepted
compared to separated markets
(425 MW)**

Consumer and Producer surplus

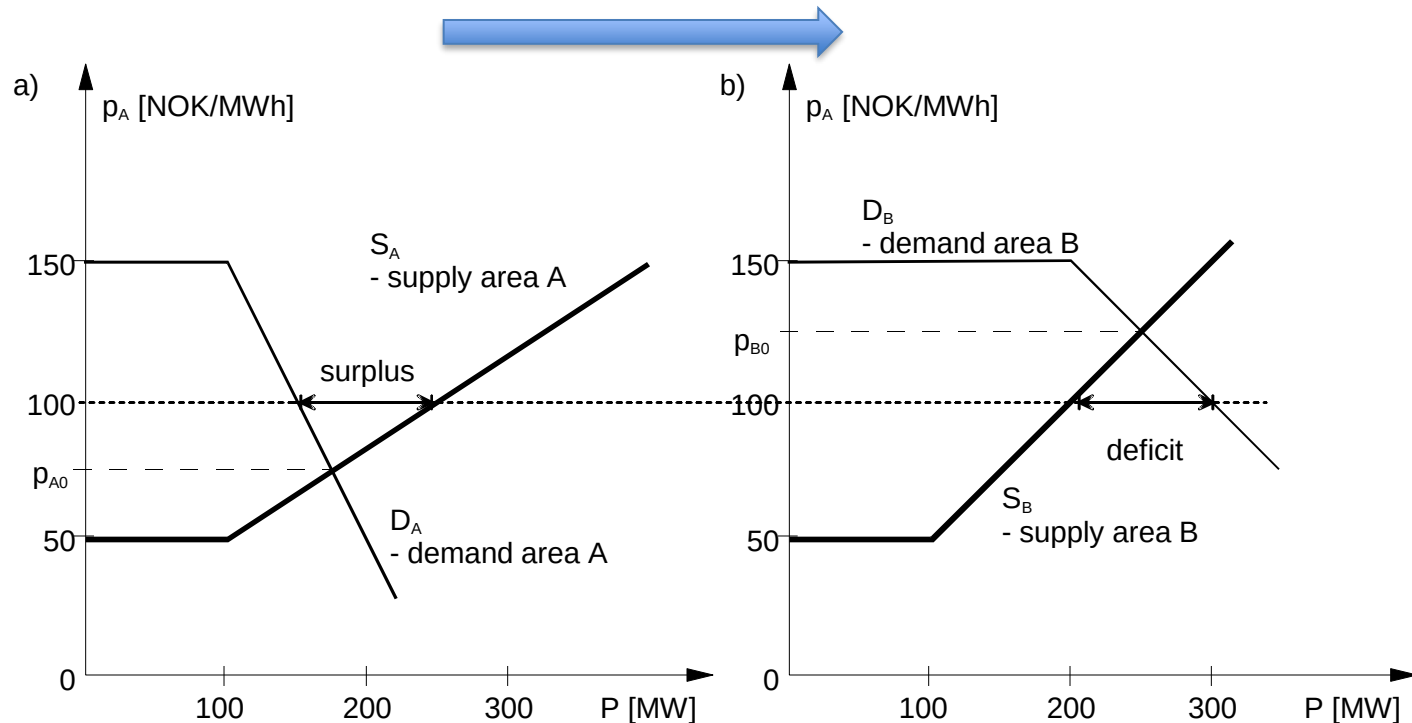


$$PS = (100 - 50) \cdot \left(200 + \frac{1}{2} \cdot 250 \right) = 16250 \text{ NOK}$$

$$CS = (150 - 100) \cdot \left(300 + \frac{1}{2} \cdot 150 \right) = 18750 \text{ NOK}$$

Total surplus: **35000** NOK

How much transfer capacity needed?



- From figure: 100 MW results in equal price
- Can also see this from demand and supply curves for each area
- With <100 MW there will be a price difference

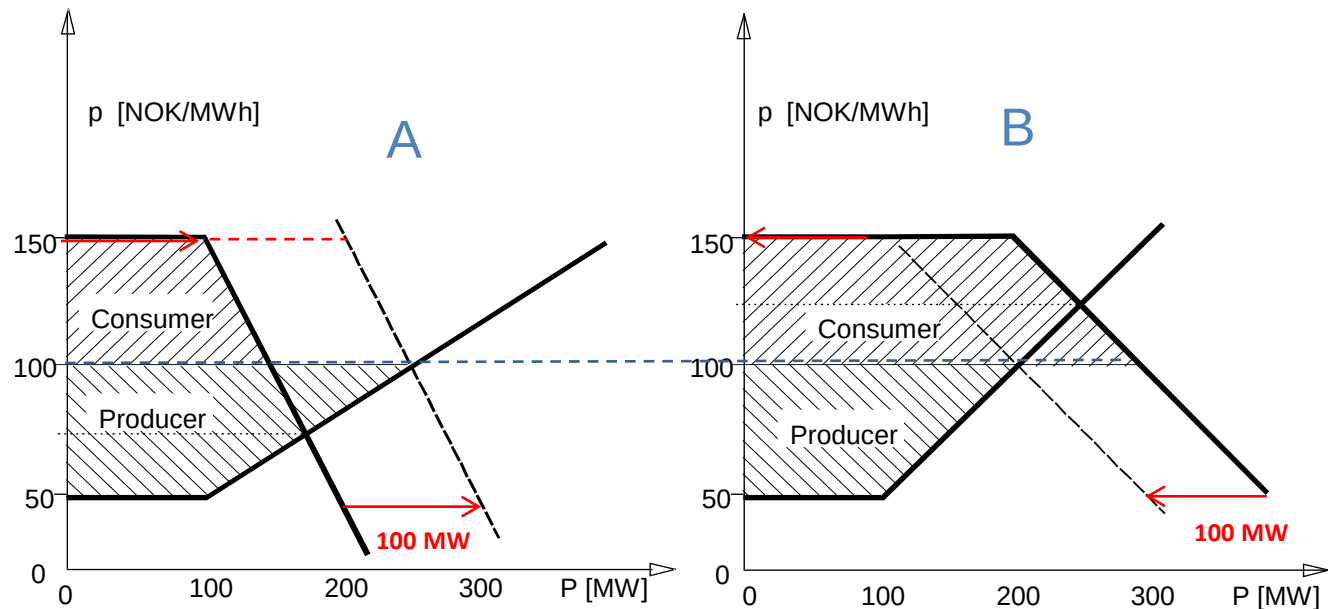
Transfer capacity viewed as shift in demand curves

$$\text{Demand: } \begin{cases} p_A = 150 & \text{for } D'_A \leq 200 \\ p_A = 350 - D'_A & \text{for } D'_A \geq 200 \end{cases}$$

$$D'_A = D_A + 100 \Rightarrow D_A = D'_A - 100$$

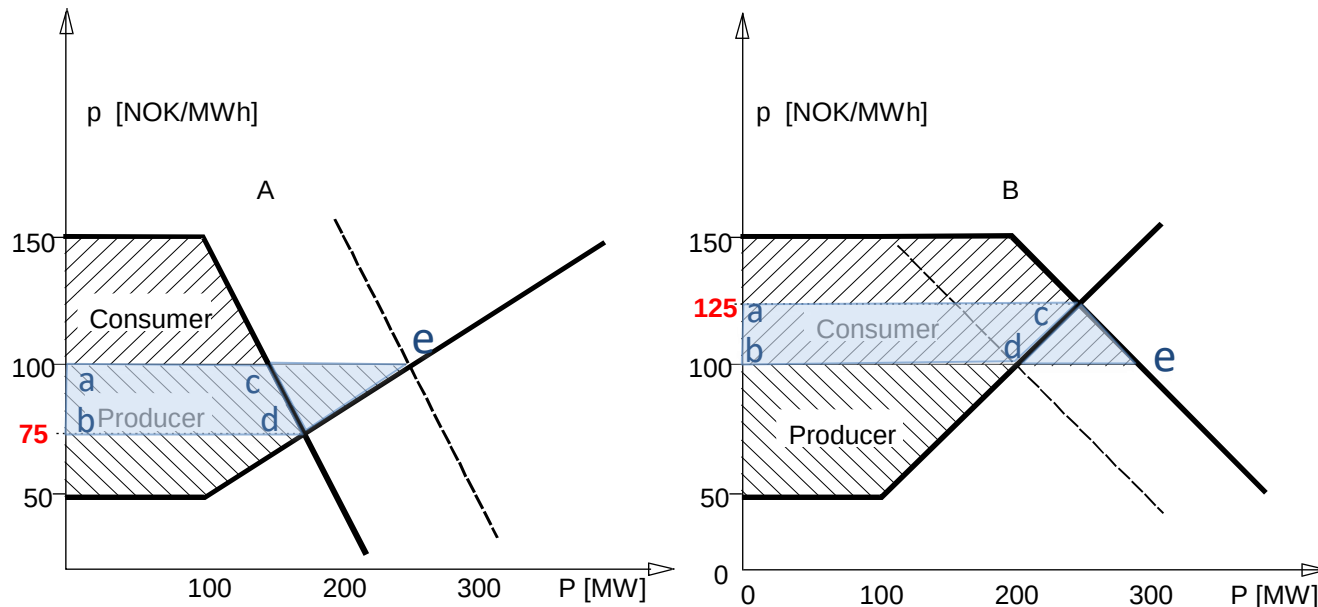
$$\text{Demand: } \begin{cases} p_B = 150 & \text{for } D'_B \leq 100 \\ p_B = 200 - 0.5D'_B & \text{for } D'_B \geq 100 \end{cases}$$

$$D'_B = D_B - 100 \Rightarrow D_B = D'_B + 100$$



- Alternatively: shift in supply curves
- Changes in producer and consumer surplus in both areas
- Increase in total surplus (social welfare) should be the criterion for the profitability of new/increased interconnection!

Impact of 100 MW transfer capacity



- Price *increase* in exporting area A
 - Increase in PS, decrease in CS
- Price *decrease* in important area B
 - Decrease in PS, increase in CS
- + Net increase in total surplus

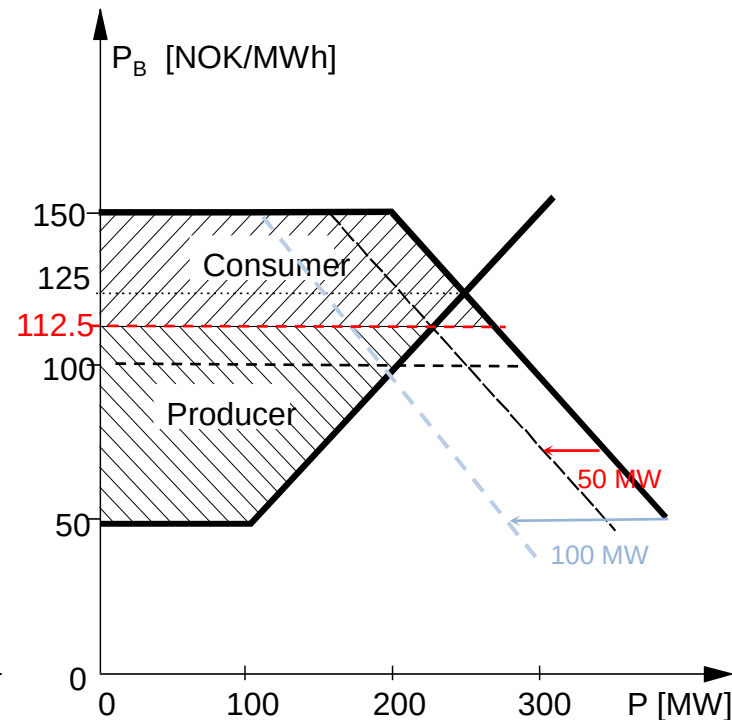
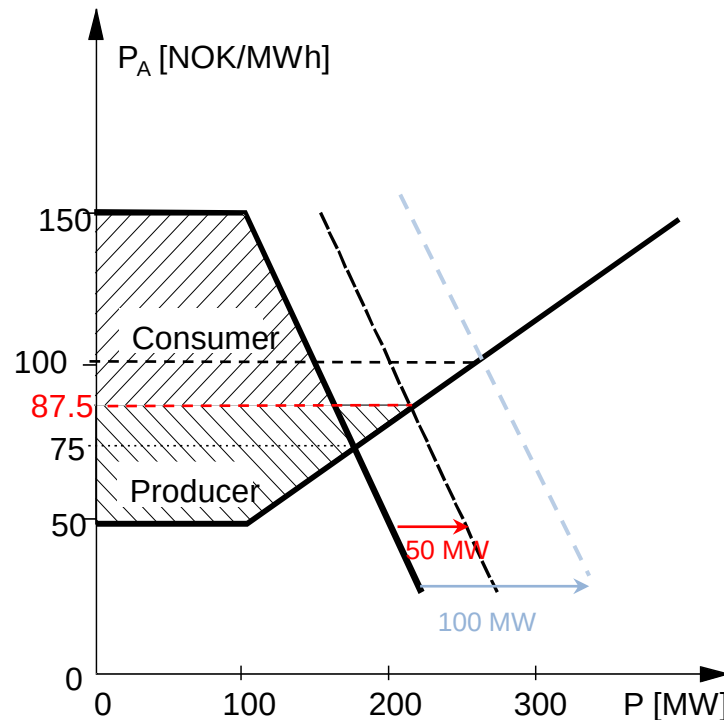
Impact of 50 MW transfer capacity

$$\text{Demand: } \begin{cases} p_A = 150 & \text{for } D'_A \leq 150 \\ p_A = 300 - D'_A & \text{for } D'_A \geq 150 \end{cases}$$

$$D'_A = D_A + 50 \Rightarrow D_A = D'_A - 50$$

$$\text{Demand: } \begin{cases} p_B = 150 & \text{for } D'_B \leq 150 \\ p_B = 225 - 0.5D'_B & \text{for } D'_B \geq 150 \end{cases}$$

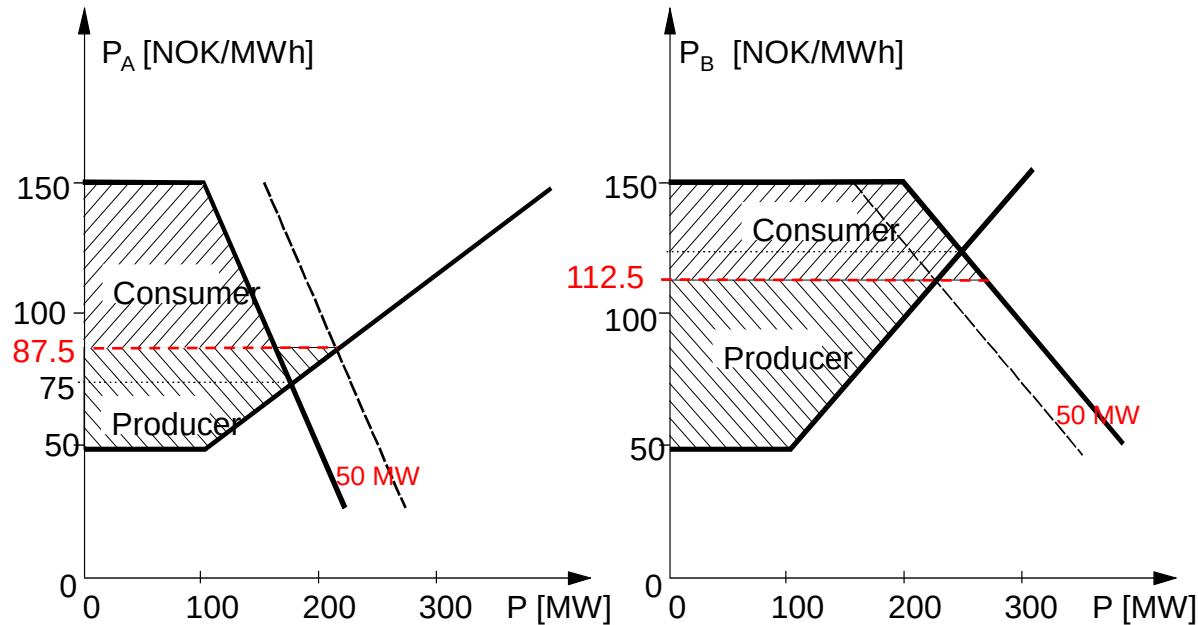
$$D'_B = D_B - 50 \Rightarrow D_B = D'_B + 50$$



$$300 - D'_A = \frac{50}{3} + \frac{1}{3} D'_A$$

Resulting in $P_B = 112.5$ and $P_A = 87.5$ (before: 75) & $D_A = 212.5 - 50 = 162.5$

Consumer and Producer surplus



$$PS_A = PS_{A0} + (87.5 - 75) \cdot (175 + \frac{1}{2} \cdot 37.5) = 5859.375 \text{ NOK}$$

$$CS_A = (150 - 87.5) \cdot (100 + \frac{1}{2} \cdot 62.5) = 8203.125 \text{ NOK}$$

$$PS_B = (112.5 - 50) \cdot (100 + \frac{1}{2} \cdot 225) = 10156.25 \text{ NOK}$$

$$CS_B = CS_{B0} + (125 - 112.5) \cdot (250 + \frac{1}{2} \cdot 25) = 8906.25 \text{ NOK}$$

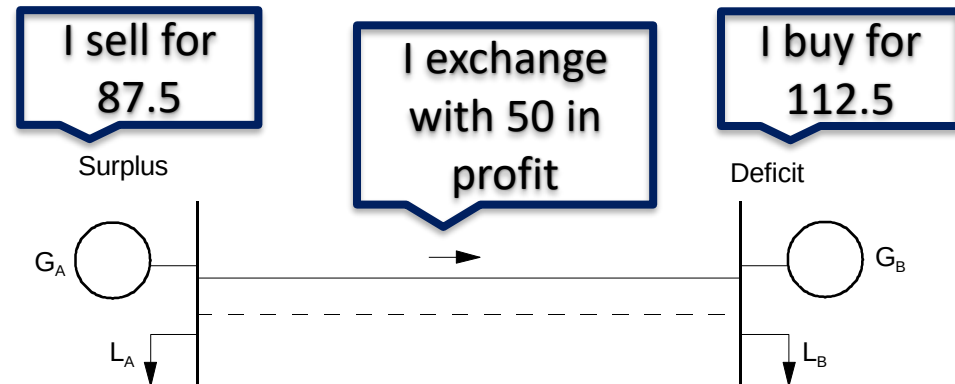
Congestion rent : transferred energy multiplied with Δp :

$$50 \cdot (112.5 - 87.5) = 1250 \text{ NOK}$$

Changes in surplus caused by increased transmission capacity (NOK)

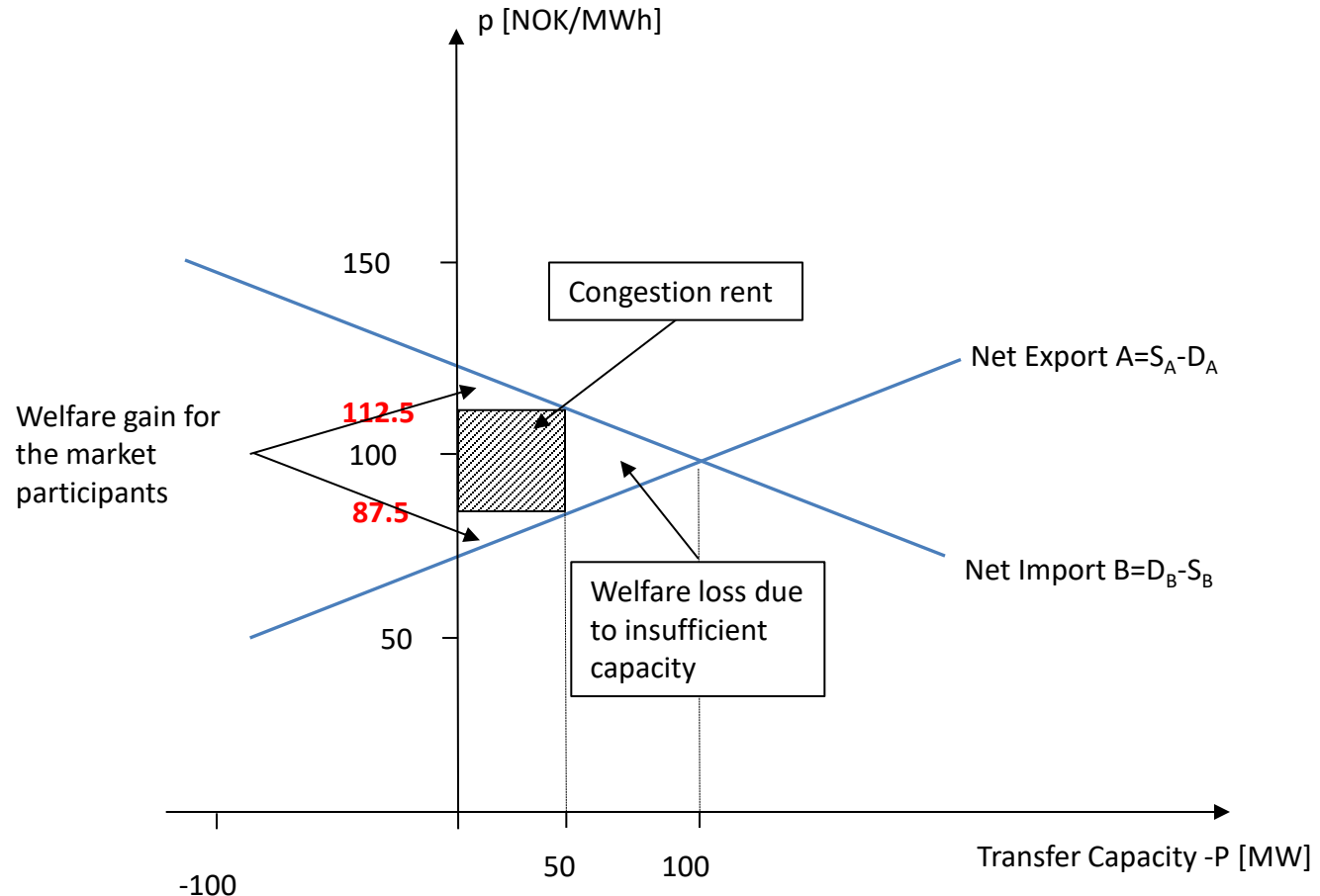
Transfer capacity (MW)	100	50
<i>Consumer surplus A</i>	-4062.5	-2109.375
<i>Producer surplus A</i>	+5312.5	+2421.875
<i>Consumer surplus B</i>	+6875	+3281.250
<i>Producer surplus B</i>	-5625	-2968.750
<i>Net benefit to consumers and producers</i>	2500	625
<i>Congestion rent</i>	0	1250
Total economic benefit	2500	1875

Congestion rent – Benefit of having different price signals



- Congestion rent:
 - Money earned from exchanging between price zones
- Income given to the grid operators
- An economic way of returning grid investment, but also worsen social welfare compared to optimal dispatch
- *“Congestion revenues that arise as a result of power exchange between countries, i.e. over interconnectors, are shared 50/50 between Statnett and our partner in the neighbouring country, whereas Statnett receives 100 per cent of revenues that arise as a result of power exchange between price areas within Norway.”*
 - Statnett: <https://www.statnett.no/en/for-stakeholders-in-the-power-industry/tariffs/congestion-revenues/>

Welfare effects of interconnection (Net Export and Import Curves)



Net Export and Import Curves

$$\text{Net Export } A = T = (S_A - D_A) = 3p_A - 50 - (250 - p_A) = -300 + 4p_A$$

$$\text{Net Import } B = T = (D_B - S_B) = 500 - 2p_B - (2p_B) = 500 - 4p_B$$

Can calculate congestion rent for capacity T:

$$S_A - D_A = T \quad -300 + 4p_A = T \quad p_A = \frac{T + 300}{4}$$

$$D_B - S_B = T \quad 500 - 4p_B = T \quad p_B = \frac{500 - T}{4}$$

$$\text{Congestion Rent}(CR): \left(\frac{500 - T}{4} - \frac{300 + T}{4} \right) \cdot T = 50T - 0.5T^2$$

Congestion rent is maximized by differentiation, find T=50

$$\frac{\partial CR}{\partial T} = 50 - T = 0 \Rightarrow T = 50 \text{ (MW)}$$

Generally maximized for $T \approx \mathbf{0.5 \text{ times market demand}}$

Net Export and Import Curves

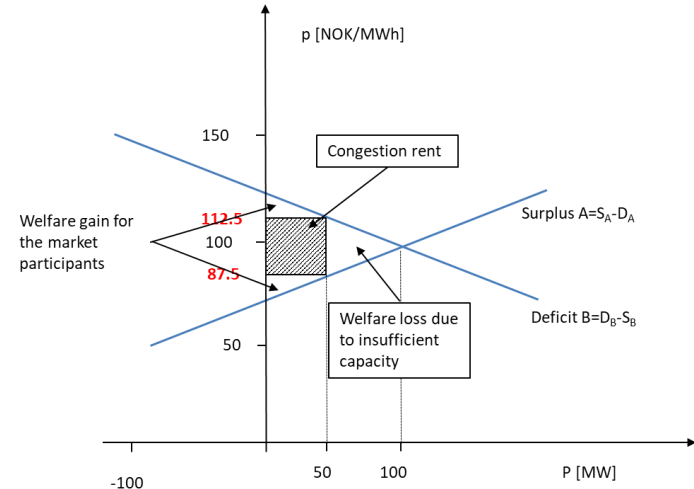
Welfare gain for the market participants

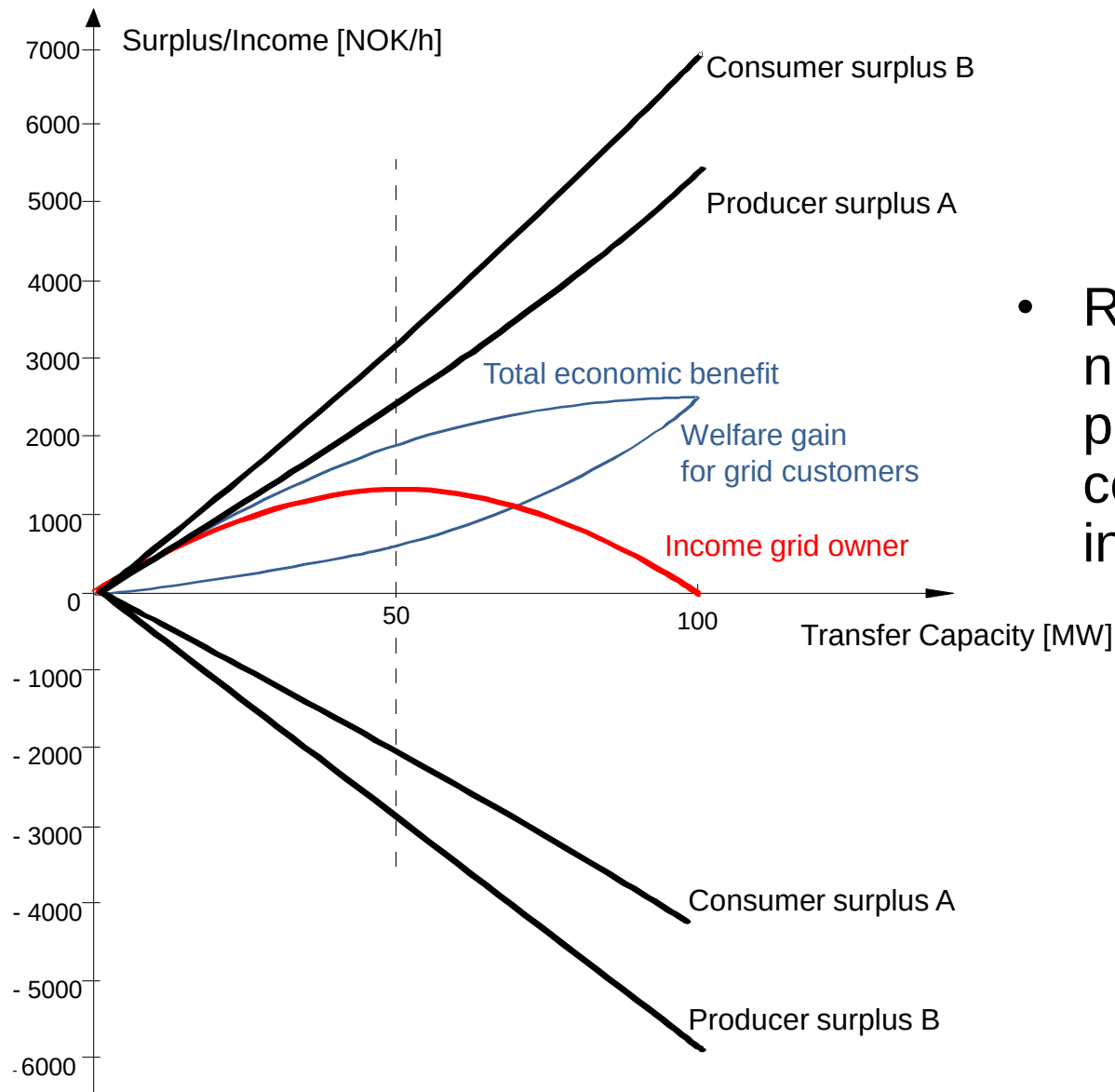
$$S_A - D_A = T \quad -300 + 4p_A = T \quad p_A = \frac{T + 300}{4}$$

$$D_B - S_B = T \quad 500 - 4p_B = T \quad p_B = \frac{500 - T}{4}$$

The area under two triangles

$$\frac{1}{2} \left(125 - \frac{500 - T}{4} \right) T + \frac{1}{2} \left(\frac{T + 300}{4} - 75 \right) T = \frac{T^2}{4}$$



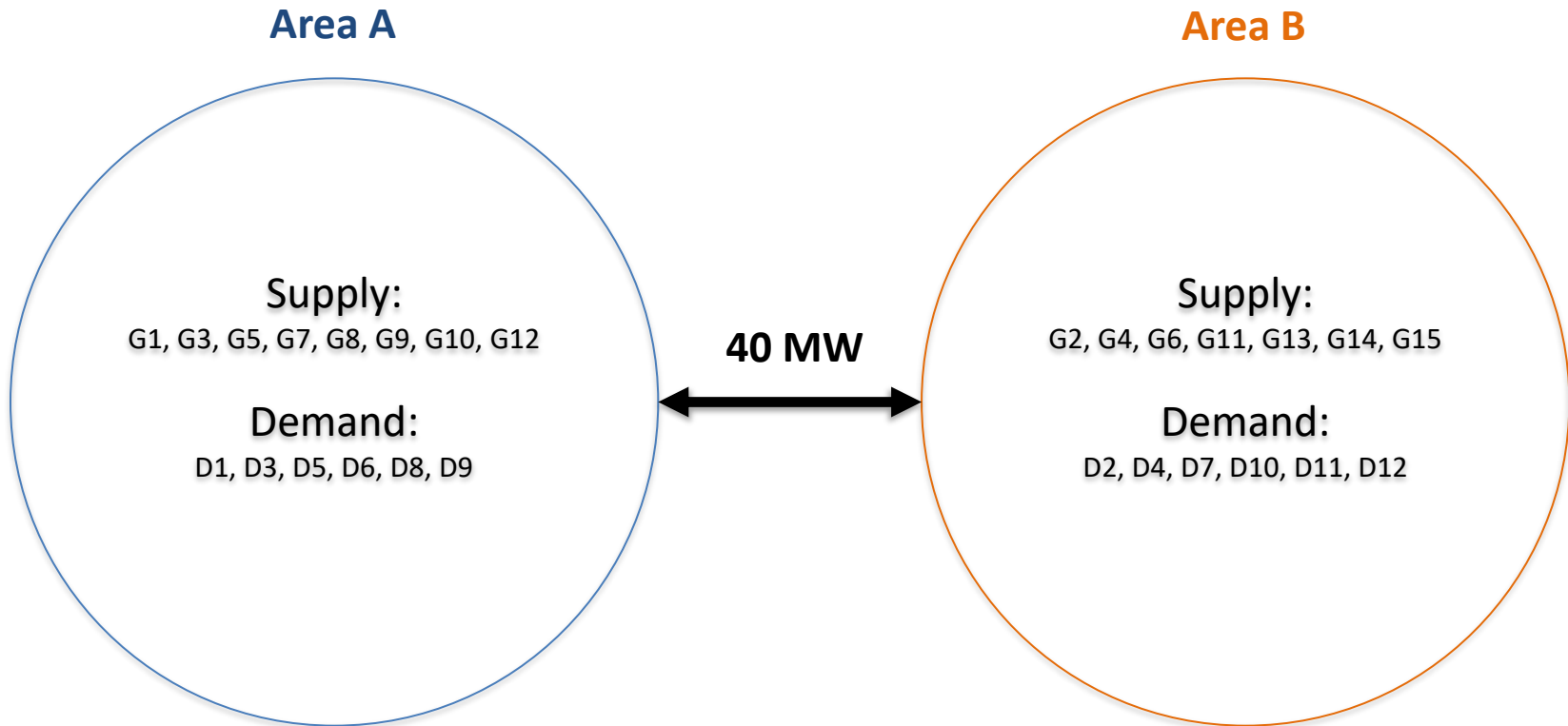


- Regulation is necessary to provide correct incentives

Area pricing as an optimization Problem

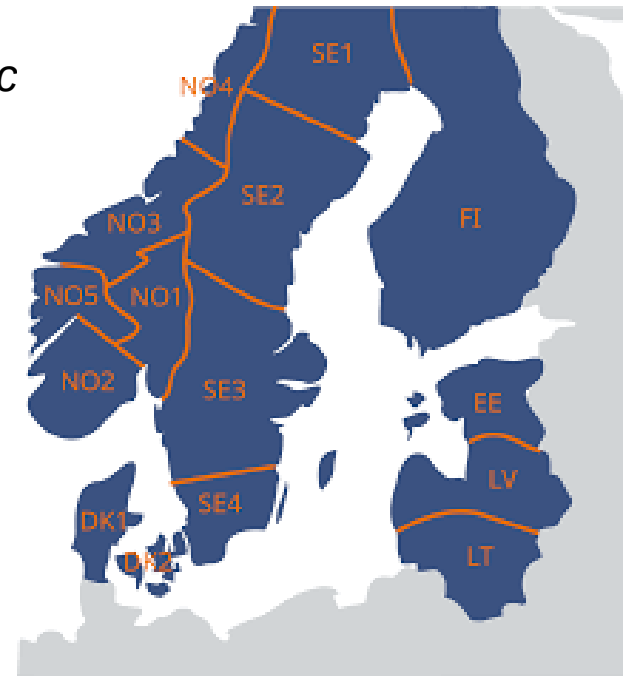
From system price to area prices

- Let us revisit our previous market clearing example,
 - considering two areas **A** and **B**



Market clearing – Area pricing

- Defining decision variables:
 - What is it we want to find values for?
 1. p_g^G - Production quantity from producer p
 2. p_d^D - Consumption quantity from consumer c
 3. T , *Exchange power between areas*
- What is the objective function?
 - Maximize social welfare (CS+PS)
- What are constraints?
 1. Power balance for each area i.e.,
 $\text{Supply} - \text{Demand} = \text{Exchange}$
 1. Supply limitation
 2. Demand limitation
 3. *Transmission limitation*



Formulating the market clearing

- The grid-constrained social welfare maximization problem can be written as:

$$\begin{aligned}
 &\text{Max}_{p_g^G, p_d^D, T} \quad SW = \underbrace{\sum_{d=1}^{N_D} U_d p_d^D}_{\text{Utility of demands}} - \underbrace{\sum_{g=1}^{N_G} C_g p_g^G}_{\text{Cost of electricity production}} \\
 &\text{subject to} \\
 &\sum_{d=1}^{N_D} p_d^{D,A} - \sum_{g=1}^{N_G} p_g^{G,A} = T \quad \rightarrow \quad \text{Power balance in area A} \\
 &\sum_{d=1}^{N_D} p_d^{D,B} - \sum_{g=1}^{N_G} p_g^{G,B} = -T \quad \rightarrow \quad \text{Power balance in area B} \\
 &0 \leq p_g^G \leq \bar{p}_g^G \quad g = 1 \dots N_G \quad \rightarrow \quad \text{Generation limits} \\
 &0 \leq p_d^D \leq \bar{p}_d^D \quad d = 1 \dots N_D \quad \rightarrow \quad \text{Consumption limits} \\
 &-40 \leq T \leq 40 \quad \rightarrow \quad \text{Transmission limits}
 \end{aligned}$$

where:

- T is the available transfer capacity of the interconnection between area A and area B

Obtaining the zonal prices

- As for the case of a single zone, the dual LP allows obtaining **market-clearing prices**
- These **2 prices** corresponds to the Lagrange multipliers for the **2 equality constraints** (i.e., Power balance equations for each area A & B):

$$\text{Max}_{p_g^G, p_d^D, T} SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G$$

subject to

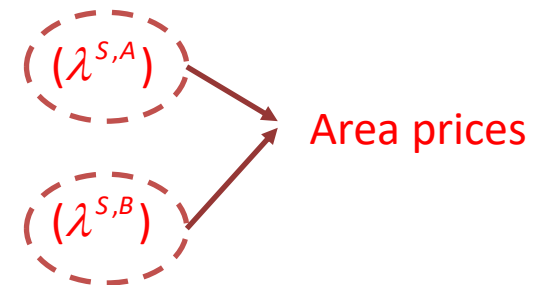
$$\sum_{d=1}^{N_D} p_d^{D,A} - \sum_{g=1}^{N_G} p_g^{G,A} = T$$

$$\sum_{d=1}^{N_D} p_d^{D,B} - \sum_{g=1}^{N_G} p_g^{G,B} = -T$$

$$0 \leq p_g^G \leq \bar{p}_g^G \quad g = 1 \dots N_G$$

$$0 \leq p_d^D \leq \bar{p}_d^D \quad d = 1 \dots N_D$$

$$-40 \leq T \leq 40$$



Localization of offers

Demand: (for a total of 1065 MWh)

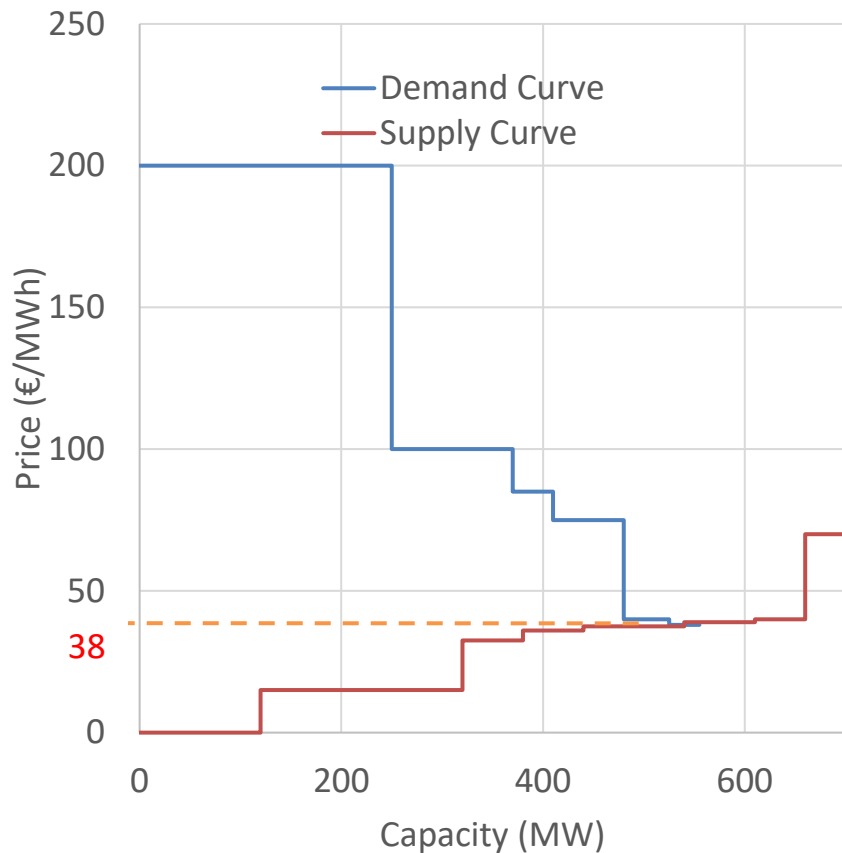
Company	id	P_d^D (MW)	U_d (€/MWh)	Area
R1	D1	250	200	A
R2	D2	300	110	B
R3	D3	120	100	A
R4	D4	80	90	B
R5	D5	40	85	A
R6	D6	70	75	A
R1	D7	60	65	B
R5	D8	45	40	A
R4	D9	30	38	A
R5	D10	35	31	B
R1	D11	25	26	B
R2	D12	10	16	B

And on the supply side

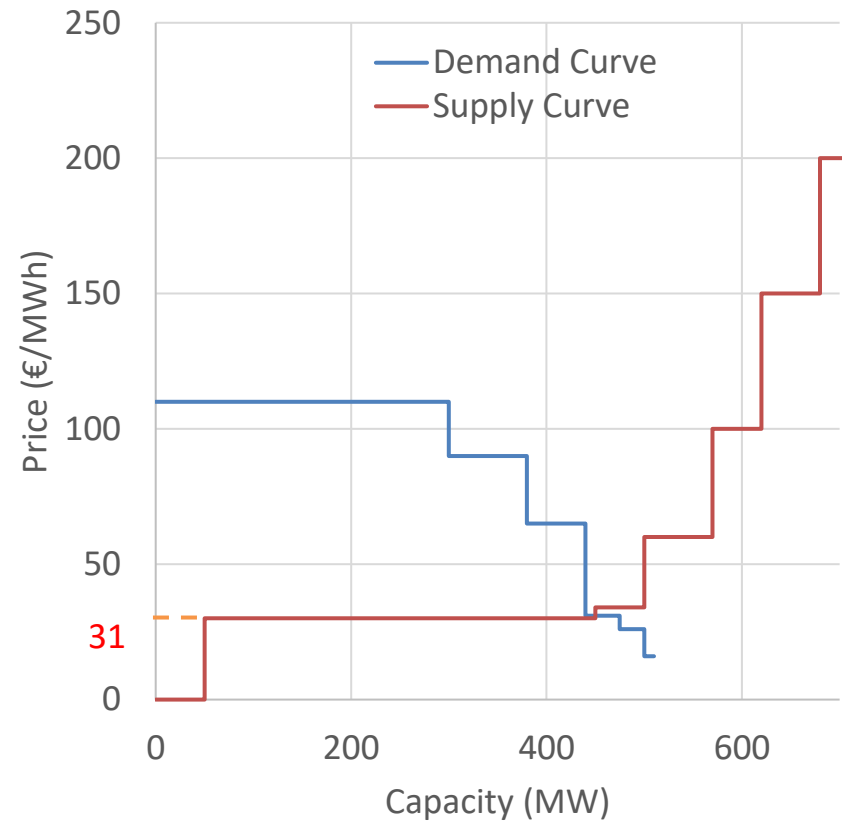
Supply: (for a total of 1435 MWh)

Company	id	P_g^G (MW)	C_g (€/MWh)	Area
C1	G1	120	0	A
C2	G2	50	0	B
C3	G3	200	15	A
C1	G4	400	30	B
C4	G5	60	32.5	A
C4	G6	50	34	B
C4	G7	60	36	A
C5	G8	100	37.5	A
C5	G9	70	39	A
C5	G10	50	40	A
C1	G11	70	60	B
C1	G12	45	70	A
C6	G13	50	100	B
C6	G14	60	150	B
C6	G15	50	200	B

Market clearing in individual areas



Total quantity=540 MW
Price= 38 €/MWh



Total quantity=450 MW
Price= 31 €/MWh

Result of optimization model (no interconnection)

By setting $T=0$, we can determine the market-clearing solution and the corresponding area prices in each individual area.

Dispatch in area "A"

```

===== Optimal Solution of Market Clearing =====
Social Welfare (SW): 94570.00

Detailed Generation and Demand by Area:

Area: A1
Generators:
Generator G01: 120.00 MW, Marginal Cost = 0.00 $/Mwh
Generator G03: 200.00 MW, Marginal Cost = 15.00 $/Mwh
Generator G05: 60.00 MW, Marginal Cost = 32.50 $/Mwh
Generator G07: 60.00 MW, Marginal Cost = 36.00 $/Mwh
Generator G08: 100.00 MW, Marginal Cost = 37.50 $/Mwh
Generator G09: 0.00 MW, Marginal Cost = 39.00 $/Mwh
Generator G10: 0.00 MW, Marginal Cost = 40.00 $/Mwh
Generator G12: 0.00 MW, Marginal Cost = 70.00 $/Mwh
Demands:
Demand D01: 250.00 MW, Marginal Utility = 200.00 $/Mwh
Demand D03: 120.00 MW, Marginal Utility = 100.00 $/Mwh
Demand D05: 40.00 MW, Marginal Utility = 85.00 $/Mwh
Demand D06: 70.00 MW, Marginal Utility = 75.00 $/Mwh
Demand D08: 45.00 MW, Marginal Utility = 40.00 $/Mwh
Demand D09: 15.00 MW, Marginal Utility = 38.00 $/Mwh

Area: A2
Generators:
Generator G02: 50.00 MW, Marginal Cost = 0.00 $/Mwh
Generator G04: 400.00 MW, Marginal Cost = 30.00 $/Mwh
Generator G06: 0.00 MW, Marginal Cost = 34.00 $/Mwh
Generator G11: 0.00 MW, Marginal Cost = 60.00 $/Mwh
Generator G13: 0.00 MW, Marginal Cost = 100.00 $/Mwh
Generator G14: 0.00 MW, Marginal Cost = 150.00 $/Mwh
Generator G15: 0.00 MW, Marginal Cost = 200.00 $/Mwh
Demands:
Demand D02: 300.00 MW, Marginal Utility = 110.00 $/Mwh
Demand D04: 80.00 MW, Marginal Utility = 90.00 $/Mwh
Demand D07: 60.00 MW, Marginal Utility = 65.00 $/Mwh
Demand D10: 10.00 MW, Marginal Utility = 31.00 $/Mwh
Demand D11: 0.00 MW, Marginal Utility = 26.00 $/Mwh
Demand D12: 0.00 MW, Marginal Utility = 16.00 $/Mwh
    
```

Dispatch in area "B"

Power Flow

Area Prices

unitary producers' surplus

unitary consumers' surplus

```

Power Flows:
Flow from A1 to A2: 0.00 MW
Flow from A2 to A1: 0.00 MW

Area Prices (Duals of Power Balance Constraints):
Area A1 price: 38.00 $/Mwh
Area A2 price: 31.00 $/Mwh

Shadow Prices for Generator Capacity Constraints:
G01: 38.00 $/Mw
G02: 31.00 $/Mw
G03: 23.00 $/Mw
G04: 1.00 $/Mw
G05: 5.50 $/Mw
G06: 0.00 $/Mw
G07: 2.00 $/Mw
G08: 0.50 $/Mw
G09: 0.00 $/Mw
G10: 0.00 $/Mw
G11: 0.00 $/Mw
G12: 0.00 $/Mw
G13: 0.00 $/Mw
G14: 0.00 $/Mw
G15: 0.00 $/Mw

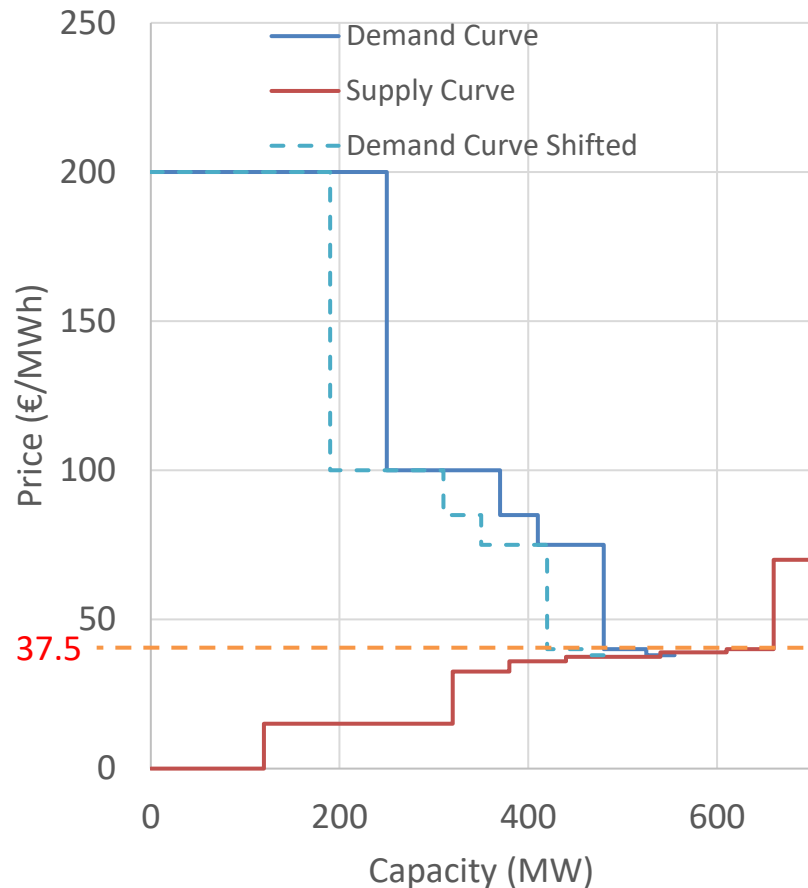
Shadow Prices for Demand Capacity Constraints:
D01: 162.00 $/Mw
D02: 79.00 $/Mw
D03: 62.00 $/Mw
D04: 59.00 $/Mw
D05: 47.00 $/Mw
D06: 37.00 $/Mw
D07: 34.00 $/Mw
D08: 2.00 $/Mw
D09: 0.00 $/Mw
D10: 0.00 $/Mw
D11: 0.00 $/Mw
D12: 0.00 $/Mw

Summary of Generation and Demand by Area:

Area: A1
Total Generation: 540.00 MW
Total Demand: 540.00 MW
Area Price: 38.00 $/Mwh

Area: A2
Total Generation: 450.00 MW
Total Demand: 450.00 MW
Area Price: 31.00 $/Mwh
    
```

Market clearing 60 MW transmission capacity

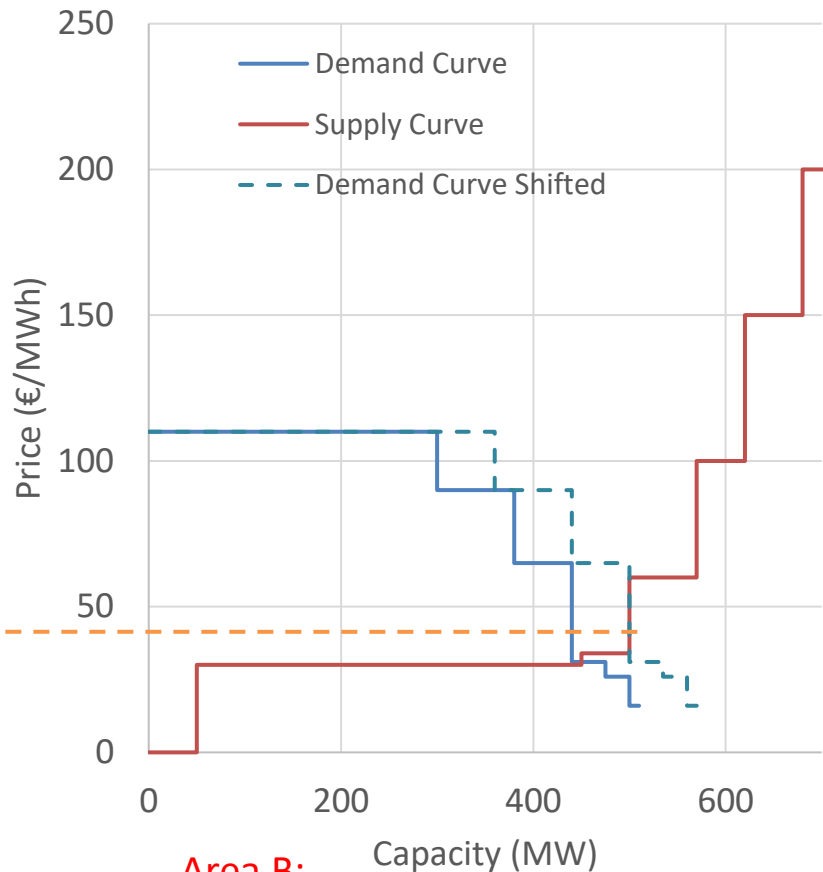


Area A:

Generation (MW)=495 MW

Demand (MW)= 555

Price= 37.5 €/MWh



Area B:

Generation (MW)=500 MW

Demand (MW)= 440

Price= 37.5 €/MWh

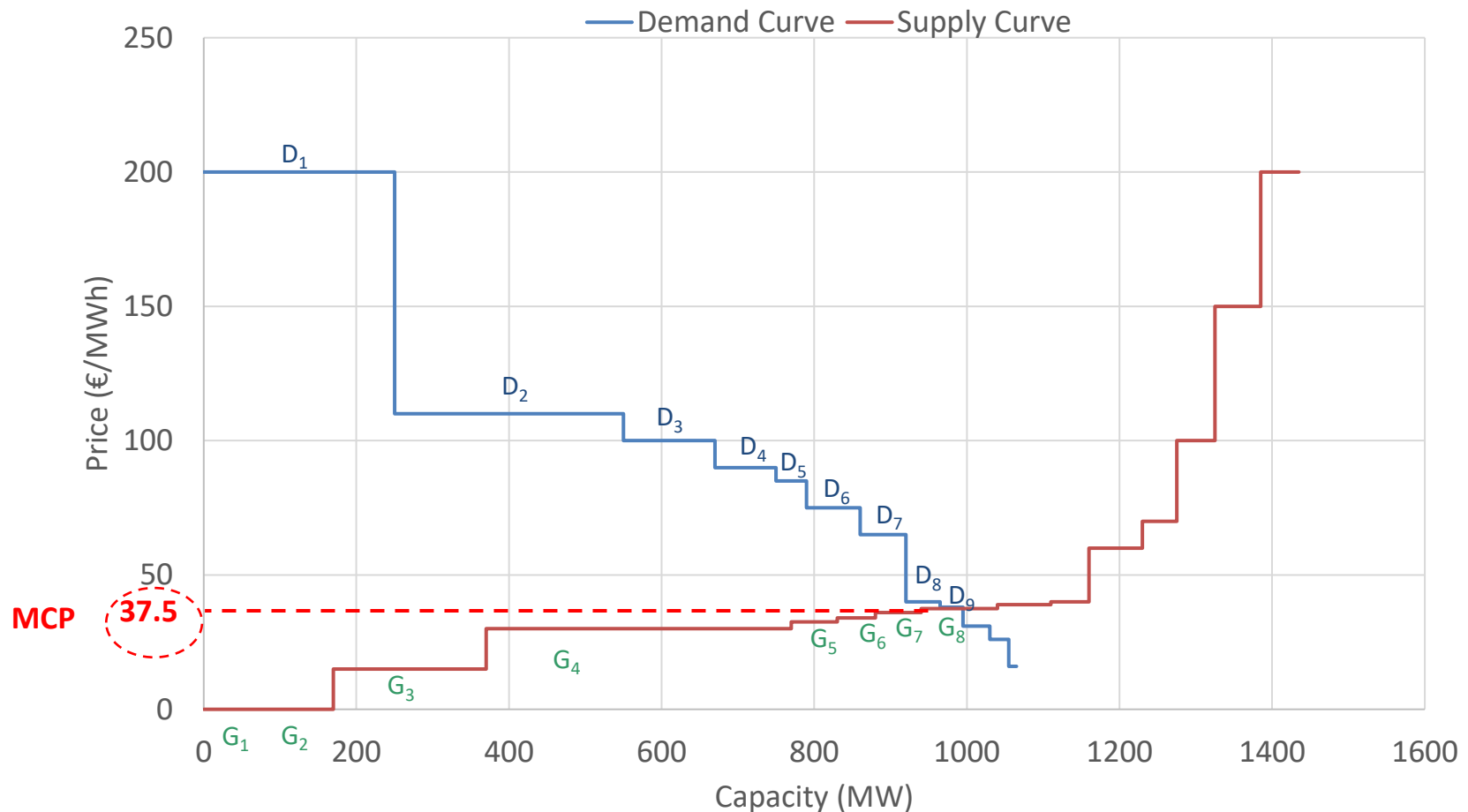
Result of optimization model (60 MW transmission capacity)

- By setting $T=60$, we can determine the market-clearing solution and the corresponding area prices in each individual area.

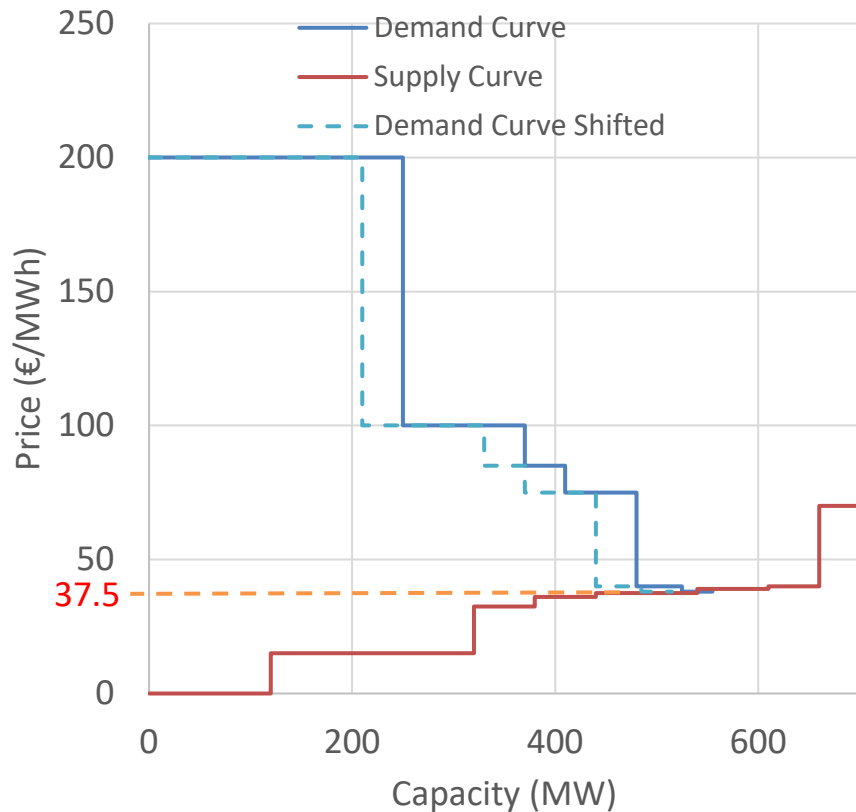
Dispatch in area "A"	===== Optimal Solution of Market Clearing ===== Social Welfare (SW): 94817.50	Power Flow	Power Flows: Flow from A1 to A2: -60.00 MW Flow from A2 to A1: 60.00 MW
	Detailed Generation and Demand by Area:	Area Prices	Area Prices (Duals of Power Balance Constraints): Area A1 price: 37.50 \$/MWh Area A2 price: 37.50 \$/MWh
Dispatch in area "B"	Area: A1 Generators: Generator G01: 120.00 MW, Marginal Cost = 0.00 \$/MWh Generator G03: 200.00 MW, Marginal Cost = 15.00 \$/MWh Generator G05: 60.00 MW, Marginal Cost = 32.50 \$/MWh Generator G07: 60.00 MW, Marginal Cost = 36.00 \$/MWh Generator G08: 55.00 MW, Marginal Cost = 37.50 \$/MWh Generator G09: 0.00 MW, Marginal Cost = 39.00 \$/MWh Generator G10: 0.00 MW, Marginal Cost = 40.00 \$/MWh Generator G12: 0.00 MW, Marginal Cost = 70.00 \$/MWh Demands: Demand D01: 250.00 MW, Marginal Utility = 200.00 \$/MWh Demand D03: 120.00 MW, Marginal Utility = 100.00 \$/MWh Demand D05: 40.00 MW, Marginal Utility = 85.00 \$/MWh Demand D06: 70.00 MW, Marginal Utility = 75.00 \$/MWh Demand D08: 45.00 MW, Marginal Utility = 40.00 \$/MWh Demand D09: 30.00 MW, Marginal Utility = 38.00 \$/MWh	unitary producers' surplus	Shadow Prices for Generator Capacity Constraints: G01: 37.50 \$/MW G02: 37.50 \$/MW G03: 22.50 \$/MW G04: 7.50 \$/MW G05: 5.00 \$/MW G06: 3.50 \$/MW G07: 1.50 \$/MW G08: 0.00 \$/MW G09: 0.00 \$/MW G10: 0.00 \$/MW G11: 0.00 \$/MW G12: 0.00 \$/MW G13: 0.00 \$/MW G14: 0.00 \$/MW G15: 0.00 \$/MW
	Area: A2 Generators: Generator G02: 50.00 MW, Marginal Cost = 0.00 \$/MWh Generator G04: 400.00 MW, Marginal Cost = 30.00 \$/MWh Generator G06: 50.00 MW, Marginal Cost = 34.00 \$/MWh Generator G11: 0.00 MW, Marginal Cost = 60.00 \$/MWh Generator G13: 0.00 MW, Marginal Cost = 100.00 \$/MWh Generator G14: 0.00 MW, Marginal Cost = 150.00 \$/MWh Generator G15: 0.00 MW, Marginal Cost = 200.00 \$/MWh Demands: Demand D02: 300.00 MW, Marginal Utility = 110.00 \$/MWh Demand D04: 80.00 MW, Marginal Utility = 90.00 \$/MWh Demand D07: 60.00 MW, Marginal Utility = 65.00 \$/MWh Demand D10: 0.00 MW, Marginal Utility = 31.00 \$/MWh Demand D11: 0.00 MW, Marginal Utility = 26.00 \$/MWh Demand D12: 0.00 MW, Marginal Utility = 16.00 \$/MWh		unitary consumers' surplus
			Summary of Generation and Demand by Area: Area: A1 Total Generation: 495.00 MW Total Demand: 555.00 MW Area Price: 37.50 \$/MWh Area: A2 Total Generation: 500.00 MW Total Demand: 440.00 MW Area Price: 37.50 \$/MWh

Merging Supply and Demand Curves

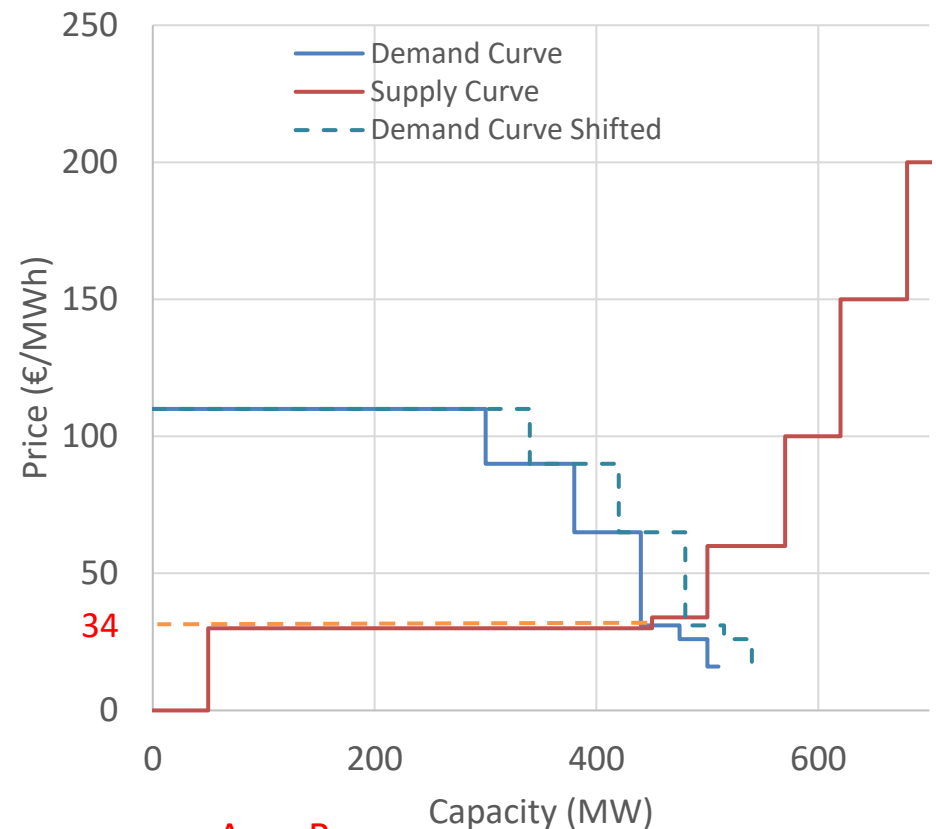
When transmission capacity is sufficiently large, the resulting equilibrium closely matches that obtained by consolidating all supply and demand into a single market (as discussed in the previous lecture).



Market clearing 40 MW transmission capacity



Area A:
 Generation (MW)=515 MW
 Demand (MW)= 555
 Price= 37.5 €/MWh



Area B:
 Generation (MW)=480 MW
 Demand (MW)= 440
 Price= 34 €/MWh

Result of optimization model (40 MW transmission capacity)

- By setting $T=40$, we can determine the market-clearing solution and the corresponding area prices in each individual area.

Dispatch in area "A"

```

===== Optimal Solution of Market Clearing =====
Social Welfare (SW): 94747.50

Detailed Generation and Demand by Area:

Area: A1
Generators:
  Generator G01: 120.00 MW, Marginal Cost = 0.00 $/MWh
  Generator G03: 200.00 MW, Marginal Cost = 15.00 $/MWh
  Generator G05: 60.00 MW, Marginal Cost = 32.50 $/MWh
  Generator G07: 60.00 MW, Marginal Cost = 36.00 $/MWh
  Generator G08: 75.00 MW, Marginal Cost = 37.50 $/MWh
  Generator G09: 0.00 MW, Marginal Cost = 39.00 $/MWh
  Generator G10: 0.00 MW, Marginal Cost = 40.00 $/MWh
  Generator G12: 0.00 MW, Marginal Cost = 70.00 $/MWh
Demands:
  Demand D01: 250.00 MW, Marginal Utility = 200.00 $/MWh
  Demand D03: 120.00 MW, Marginal Utility = 100.00 $/MWh
  Demand D05: 40.00 MW, Marginal Utility = 85.00 $/MWh
  Demand D06: 70.00 MW, Marginal Utility = 75.00 $/MWh
  Demand D08: 45.00 MW, Marginal Utility = 40.00 $/MWh
  Demand D09: 30.00 MW, Marginal Utility = 38.00 $/MWh

Area: A2
Generators:
  Generator G02: 50.00 MW, Marginal Cost = 0.00 $/MWh
  Generator G04: 400.00 MW, Marginal Cost = 30.00 $/MWh
  Generator G06: 30.00 MW, Marginal Cost = 34.00 $/MWh
  Generator G11: 0.00 MW, Marginal Cost = 60.00 $/MWh
  Generator G13: 0.00 MW, Marginal Cost = 100.00 $/MWh
  Generator G14: 0.00 MW, Marginal Cost = 150.00 $/MWh
  Generator G15: 0.00 MW, Marginal Cost = 200.00 $/MWh
Demands:
  Demand D02: 300.00 MW, Marginal Utility = 110.00 $/MWh
  Demand D04: 80.00 MW, Marginal Utility = 90.00 $/MWh
  Demand D07: 60.00 MW, Marginal Utility = 65.00 $/MWh
  Demand D10: 0.00 MW, Marginal Utility = 31.00 $/MWh
  Demand D11: 0.00 MW, Marginal Utility = 26.00 $/MWh
  Demand D12: 0.00 MW, Marginal Utility = 16.00 $/MWh
  
```

Dispatch in area "B"

Power Flow

Area Prices

unitary producers' surplus

unitary consumers' surplus

```

Power Flows:
  Flow from A1 to A2: -40.00 MW
  Flow from A2 to A1: 40.00 MW

Area Prices (Duals of Power Balance Constraints):
  Area A1 price: 37.50 $/MWh
  Area A2 price: 34.00 $/MWh

Shadow Prices for Generator Capacity Constraints:
  G01: 37.50 $/MW
  G02: 34.00 $/MW
  G03: 22.50 $/MW
  G04: 4.00 $/MW
  G05: 5.00 $/MW
  G06: 0.00 $/MW
  G07: 1.50 $/MW
  G08: 0.00 $/MW
  G09: 0.00 $/MW
  G10: 0.00 $/MW
  G11: 0.00 $/MW
  G12: 0.00 $/MW
  G13: 0.00 $/MW
  G14: 0.00 $/MW
  G15: 0.00 $/MW
  
```

```

Shadow Prices for Demand Capacity Constraints:
  D01: 162.50 $/MW
  D02: 76.00 $/MW
  D03: 62.50 $/MW
  D04: 56.00 $/MW
  D05: 47.50 $/MW
  D06: 37.50 $/MW
  D07: 31.00 $/MW
  D08: 2.50 $/MW
  D09: 0.50 $/MW
  
```

Results for our auction example

- Results are the same than those based on the import-export approach

Area A



- Supply side: {G1, G3, G5, G7, G8} (but only 75 MWh for G8)-Total: 515 MWh
 - Demand side: {D1, D3, D5, D6, D8, D9}-Total: 555 MWh
- **Zonal price: 37.5 €/MWh**

Area B



- Supply side: {G2, G4, G6} (but only 30 MWh for G6)-Total: 480 MWh
 - Demand side: {D2, D4, D7}-Total: 440 MWh
- **Zonal price: 34 €/MWh**

- However, all zones are modeled at once, and the approach can scale readily

Accepted Bids and offers

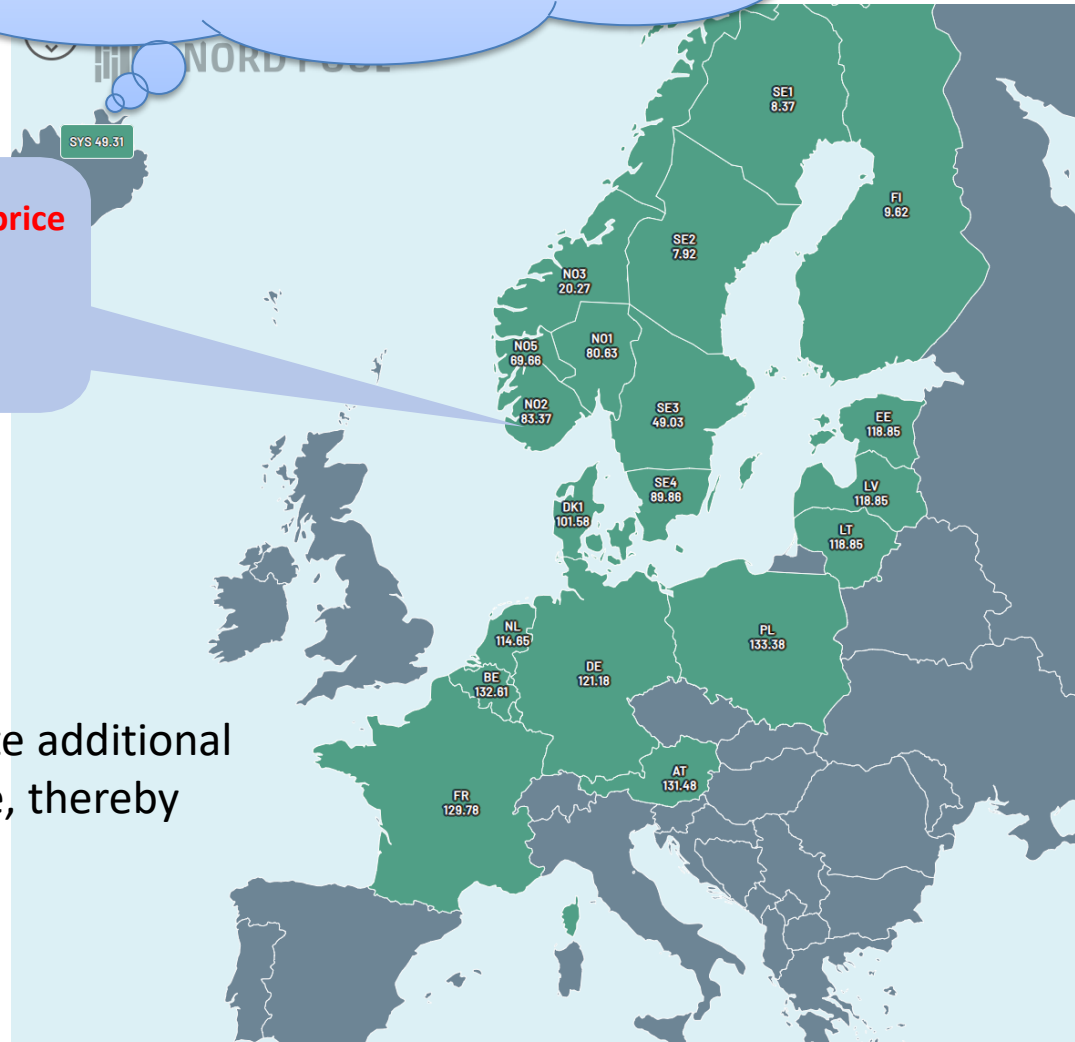
id	P_d^D (MW)	U_d (€/MWh)	Area	Total
D1	250	200	A	555
D3	120	100	A	
D5	40	85	A	
D6	70	75	A	
D8	45	40	A	
D9	30	38	A	
D2	300	110	B	440
D4	80	90	B	
D7	60	65	B	
D10	35	31	B	
D11	25	26	B	
D12	10	16	B	

id	P_g^G (MW)	C_g (€/MWh)	Area	Total
G1	120	0	A	515
G3	200	15	A	
G5	60	32.5	A	
G7	60	36	A	
G8	75(100)	37.5	A	
G9	70	39	A	
G10	50	40	A	480
G12	45	70	A	
G2	50	0	B	
G4	400	30	B	
G6	30 (50)	34	B	
G11	70	60	B	
G13	50	100	B	
G14	60	150	B	
G15	50	200	B	

NordPool

The system price corresponds to the market-clearing price (MCP) obtained by combining the aggregate supply and demand curves.

The grid limitations cause the price difference in associated bidding zones



It is straightforward to incorporate additional areas into the preceding example, thereby scaling up the solution.

Optimal exchange capacity?

- Society:
 - A correct incentive to invest should reflect the economic benefit to society (welfare gain) as a result of increased capacity
- Producers in surplus area and consumers in deficit area:
 - So much that $\Delta \text{price} = 0$
- Producers in deficit area and consumers in surplus area:
 - None at all
- TSO?
 - Or the party that receives the congestion rent

Area pricing - final remarks

- Revenue to TSO
- Often used as argument against area pricing
- TSO has incentive to set capacity at 50 % of market demand
- Important: **regulation**
- Statnett must pay back all congestion rent to grid users through lower grid tariffs

Statnett and congestion rent - 2022

- In 2022, Statnett earned 2.2 billion Euro in congestion rent
 - 0.57 billion Euro in 2021!
- Earnings were balanced between internal differences and exchange with interconnectors
- Statnett is a regulated monopoly
 - Decreased grid tariff cost to end-users to counter the increased revenue (1.1 billion Euro)

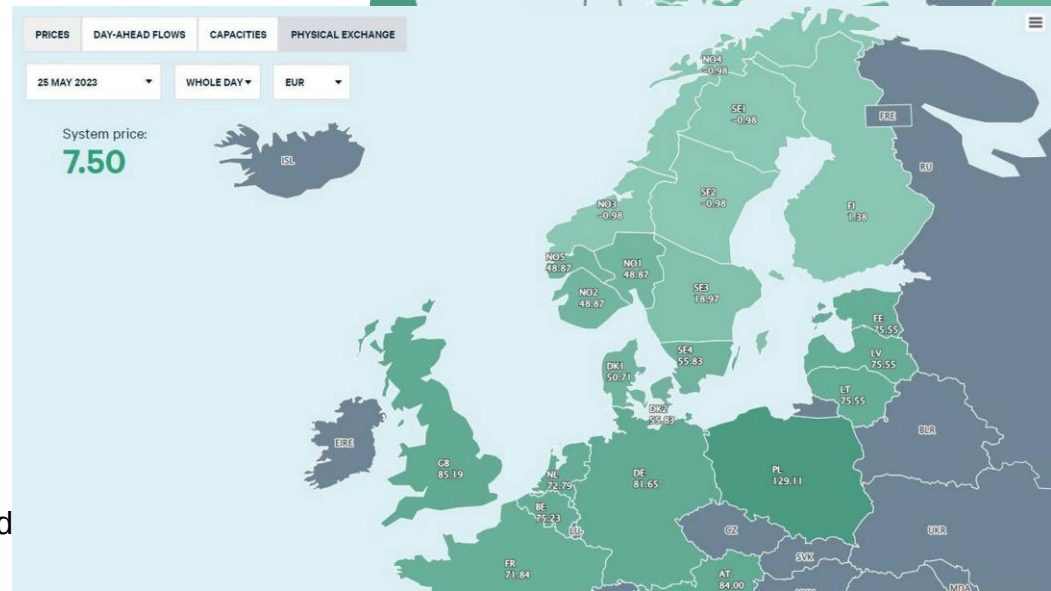
Flaskehalsinntekter 2022 - mill. EUR -pr. land

	Norge Internt	NO-SE	NO-DK (Skagerak 1-4)	NO-NL (NorNed)	NO-DE (NordLink)	NO-UK (NSL)	SUM	Akk. hittil i år
Januar	59,8	15,6	13,4	11,6	18,7	20,8	140,0	140,0
Februar	35,4	17,8	11,7	9,5	9,9	22,9	107,1	247,1
Mars	60,9	24,1	30,7	17,0	38,5	37,8	208,9	456,1
April	58,6	39,1	12,4	5,2	12,8	8,3	136,3	592,4
Mai	61,5	27,2	5,0	0,2	7,5	6,1	107,6	699,9
Juni	98,4	20,5	16,7	0,0	16,5	3,7	155,8	855,8
Juli	224,2	27,9	13,6	0,0	19,6	12,7	297,9	1 153,7
August	304,9	35,6	9,6	0,0	10,1	6,4	366,7	1 520,4
September	177,2	61,3	10,5	0,0	9,7	13,2	271,9	1 792,3
Oktober	56,5	15,4	5,0	0,7	10,4	7,9	95,8	1 888,1
November	31,6	35,5	11,5	15,0	20,0	20,6	134,2	2 022,3
Desember	65,7	45,2	10,8	4,5	8,3	10,9	145,3	2 167,6
SUM	1 234,7	364,9	150,8	63,7	182,1	171,4	2 167,6	

<https://www.statnett.no/om-statnett/nyheter-og-pressemedlinger/nyhetsarkiv-2023/rekordhoye-flaskehalsinntekter-i-2022/>

Interconnectors – how are prices influenced by each other?

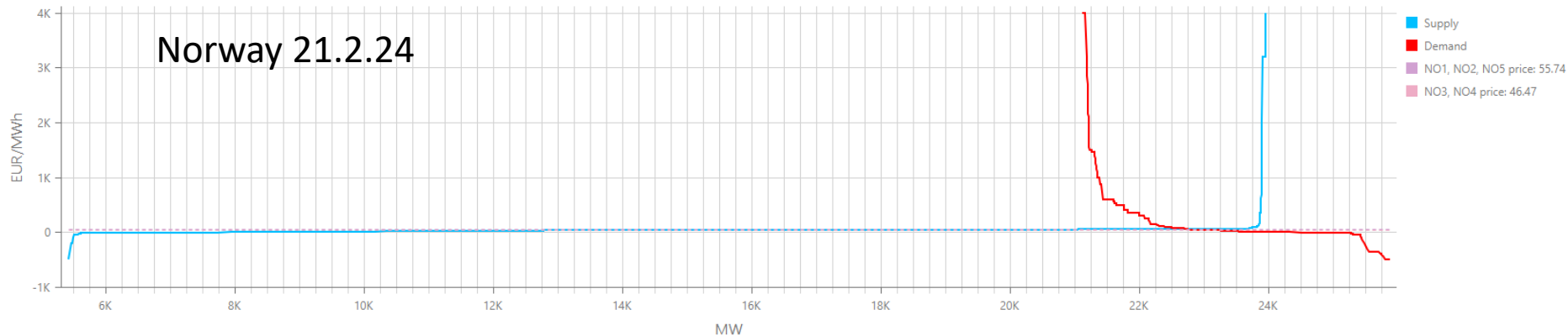
- Norway is connected to many European countries
 - By neighboring countries
 - Or through interconnectors
- If there is sufficient transmission, prices are equal for all price zones
- If there is insufficient capacity, area prices are formed



Interconnectors – how are prices influenced by each other?

Delivery period: 17:00 - 18:00 (CET)

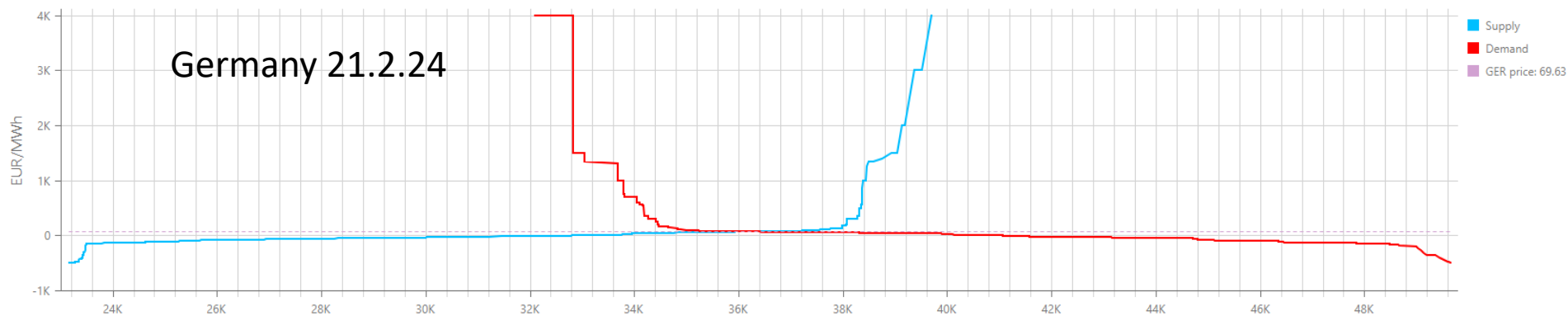
Norway 21.2.24



Transmission between: 670 MWh to Germany

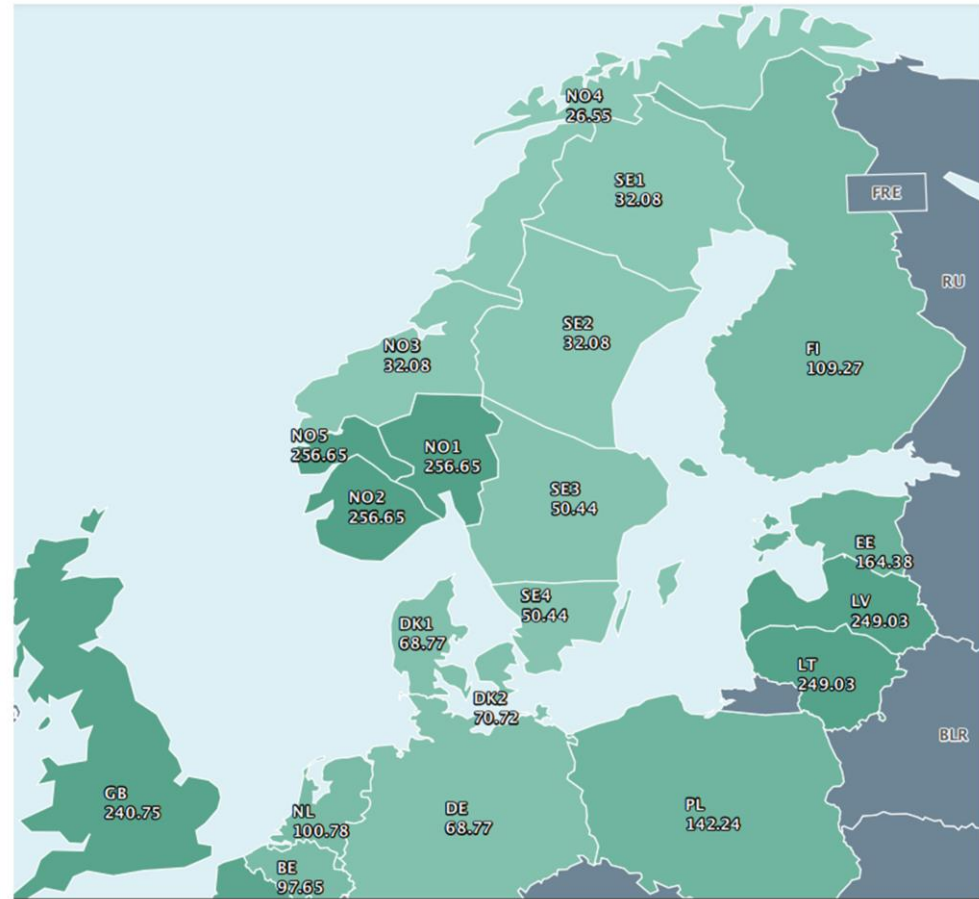
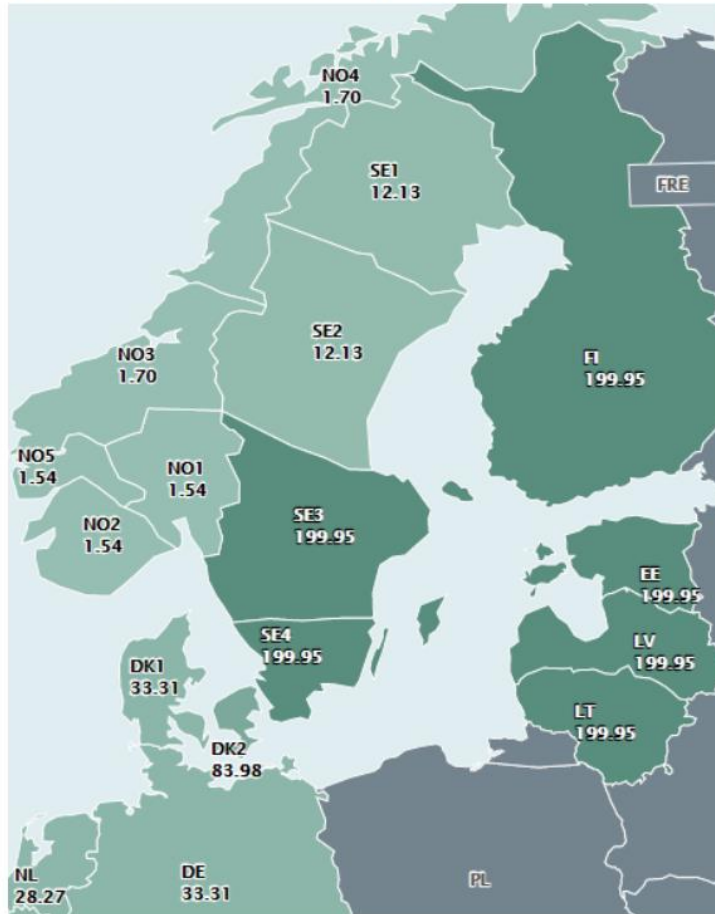
Delivery period: 17:00 - 18:00 (CET)

Germany 21.2.24



Interconnectors can have high impact on high-price areas (what if there were no cables?)

NORDIC AREA PRICES, 25 JUNE 2020 H12-13



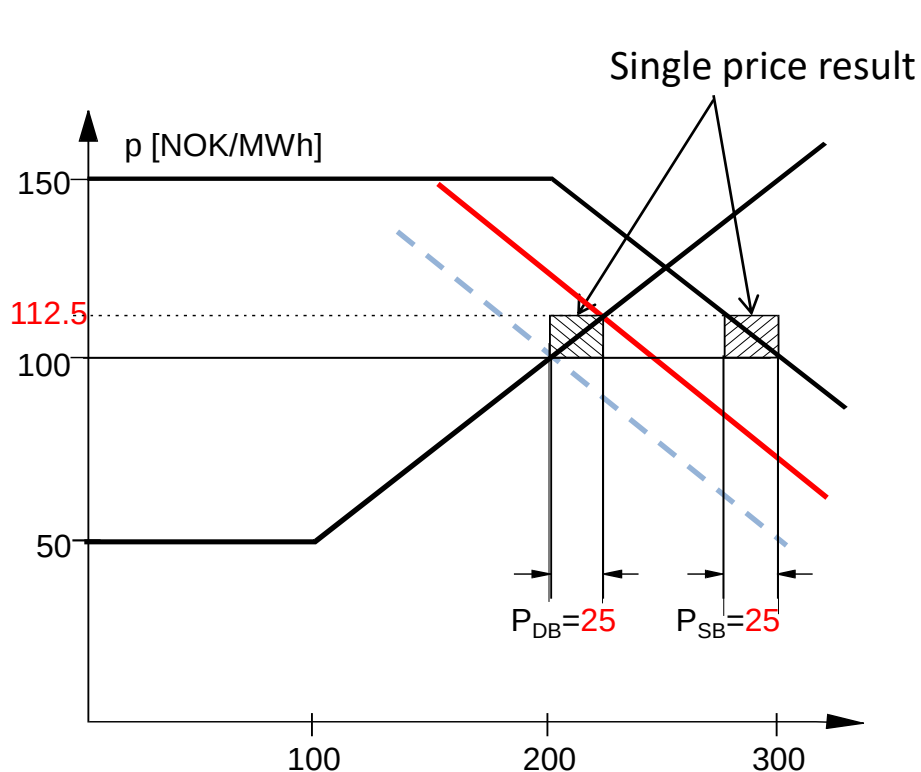
Buy-back or Counter trade

- Area pricing leads to (potentially large) price differences
 - Reference price for financial markets
 - Sometimes at a large cost to consumers (deficit area)
 - ... or producers (e.g. area with high wind production)
 - Market power
-
- Counter trade is often used as an alternative to relieve congestion
 - Or in addition to area pricing ("internal" congestion)

Buy-back or Counter trade

- Active involvement from System Operator's (SO's) side
- SO buys on the deficit side
- SO sells on surplus side
- SO suffers a net loss on the transaction because it buys at a higher price than it sells

Buy-back, uniform pricing, area B, 50 MW transfer cap

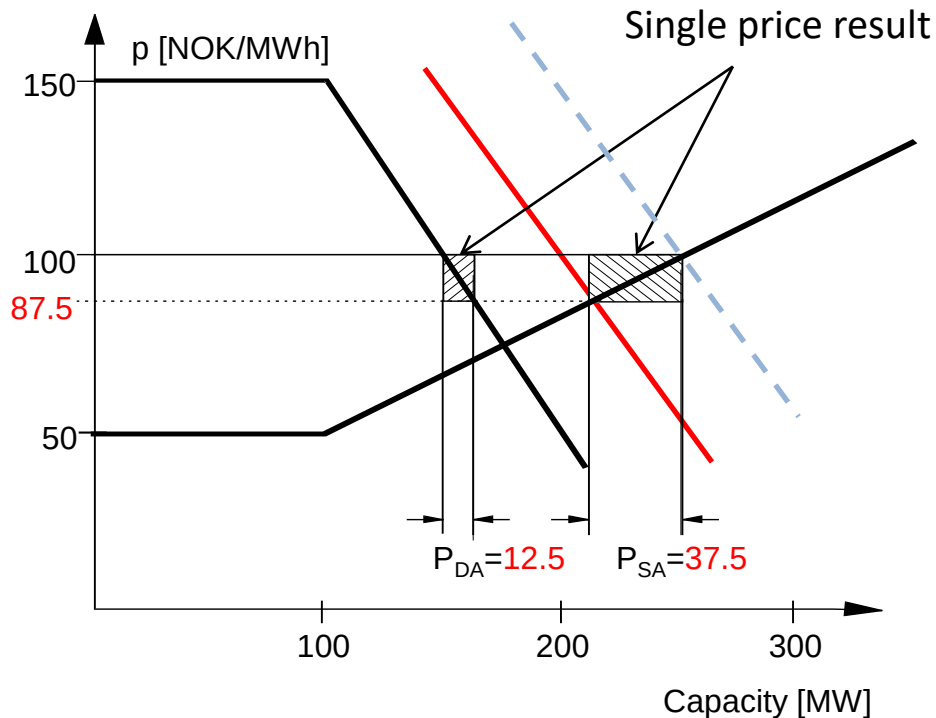


	Sysetm Price	Buy-back Price
Single price	100 →	112.5
$D_A =$	300 →	275
$S_A =$	200 →	225

Must increase supply with 25 MW and decrease demand with 25 MW, costs:

$$(112.5 - 100) \cdot (25 + 25) = 625 \text{ NOK}$$

Buy-back, uniform pricing, area A, 50 MW transfer cap



Sysetm Price Buy-back Price

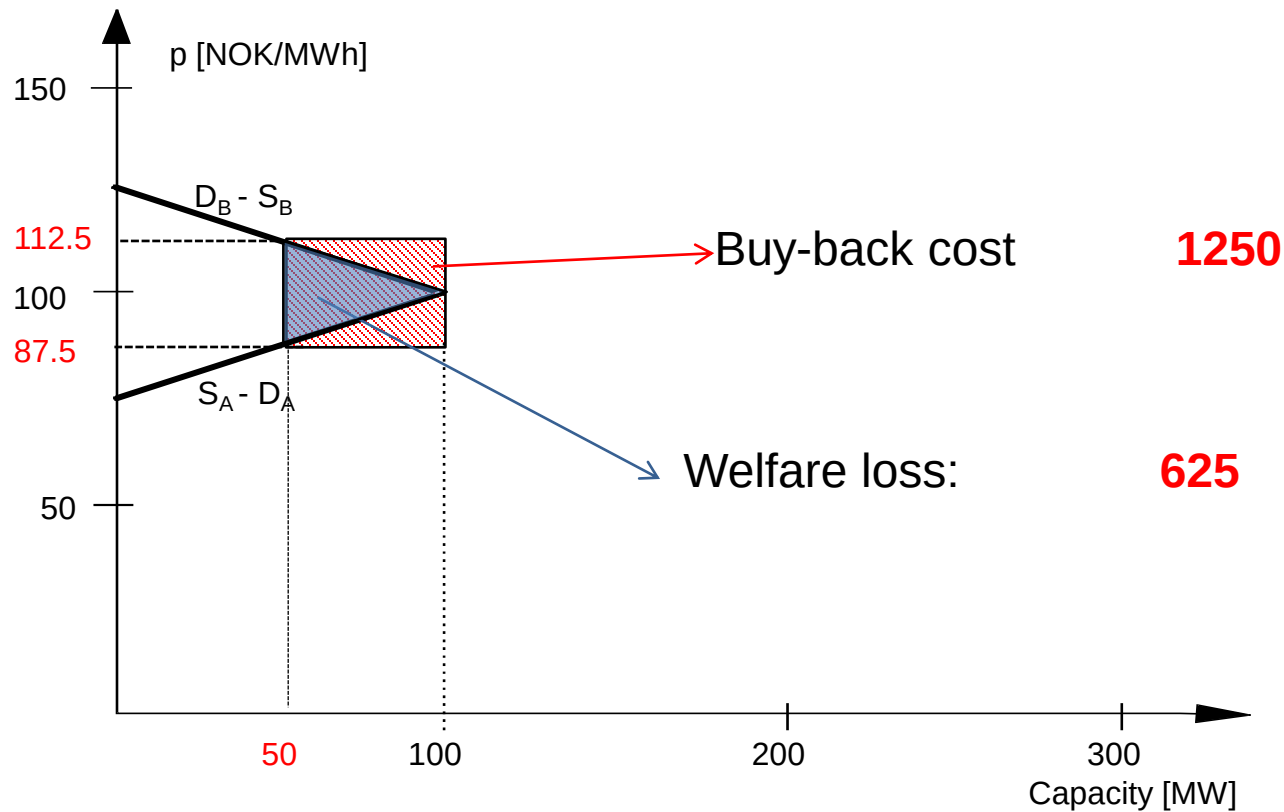
Single price 100 $\rightarrow$ 87.5
 $D_A =$ 150 $\rightarrow$ 162.5
 $S_A =$ 250 $\rightarrow$ 212.5

Must increase demand
 with 12.5 MW and
decrease supply with
 37.5 MW, costs:

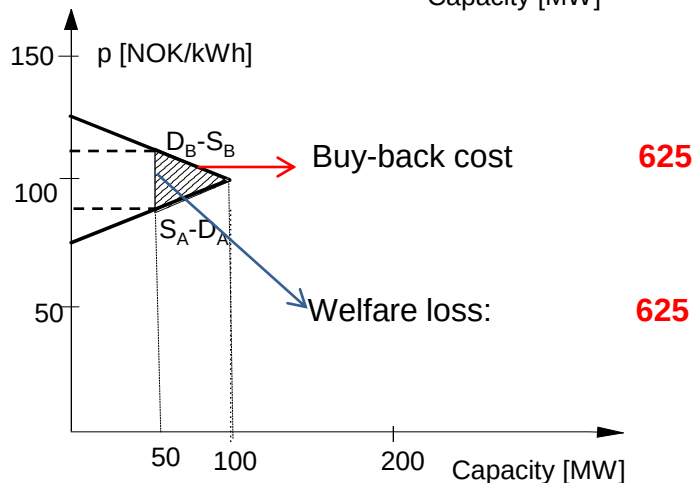
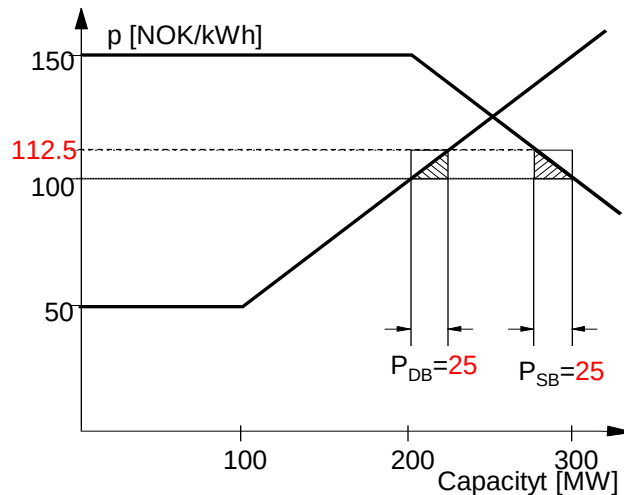
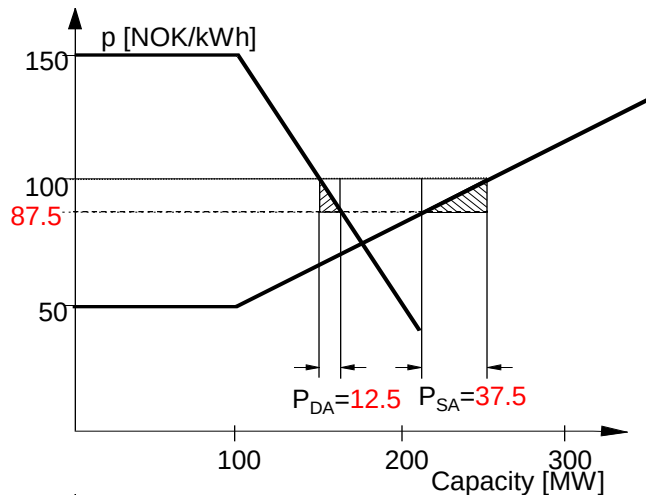
$$(100 - 87.5) \cdot (12.5 + 37.5) = 625 \text{ NOK}$$

In reality often only
 production

Buy-back (Net Export and Import Curves)



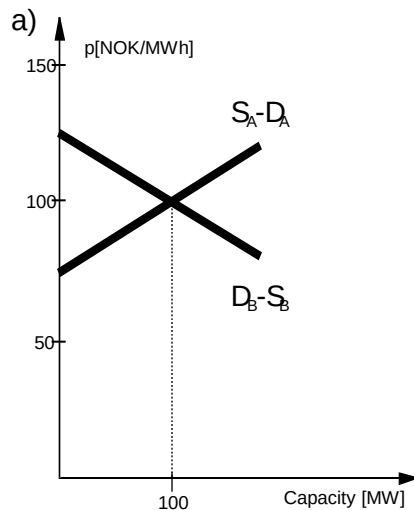
Buy-back with price discrimination (Pay-as-bid)



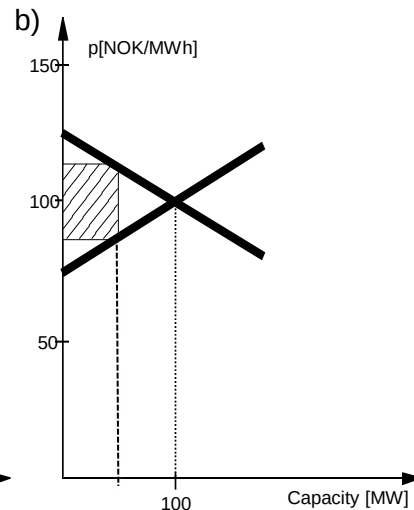
Pay-as-bid vs Uniform price

- PAB is apparently cheaper for the TSO than uniform price
- However: market actors will adapt their offers/bids
 - If they offer marginal cost, they do not make any profits
- Therefore they try to estimate the final market price and adapt their bid to that

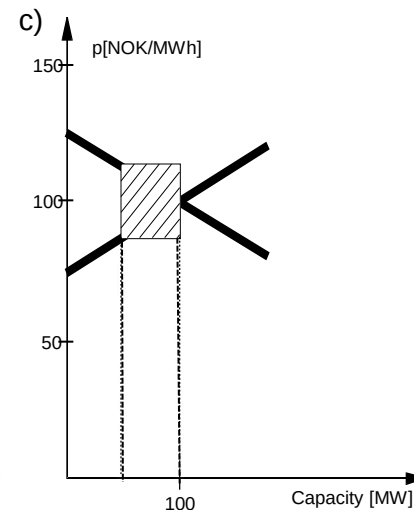
Congestion management from the TSO point-of-view



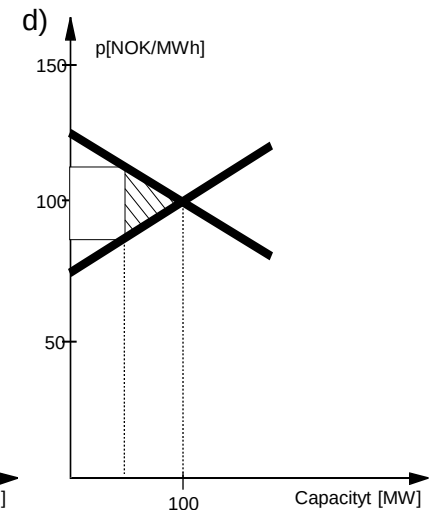
No constraint



Area pricing
congestion rent
= revenue



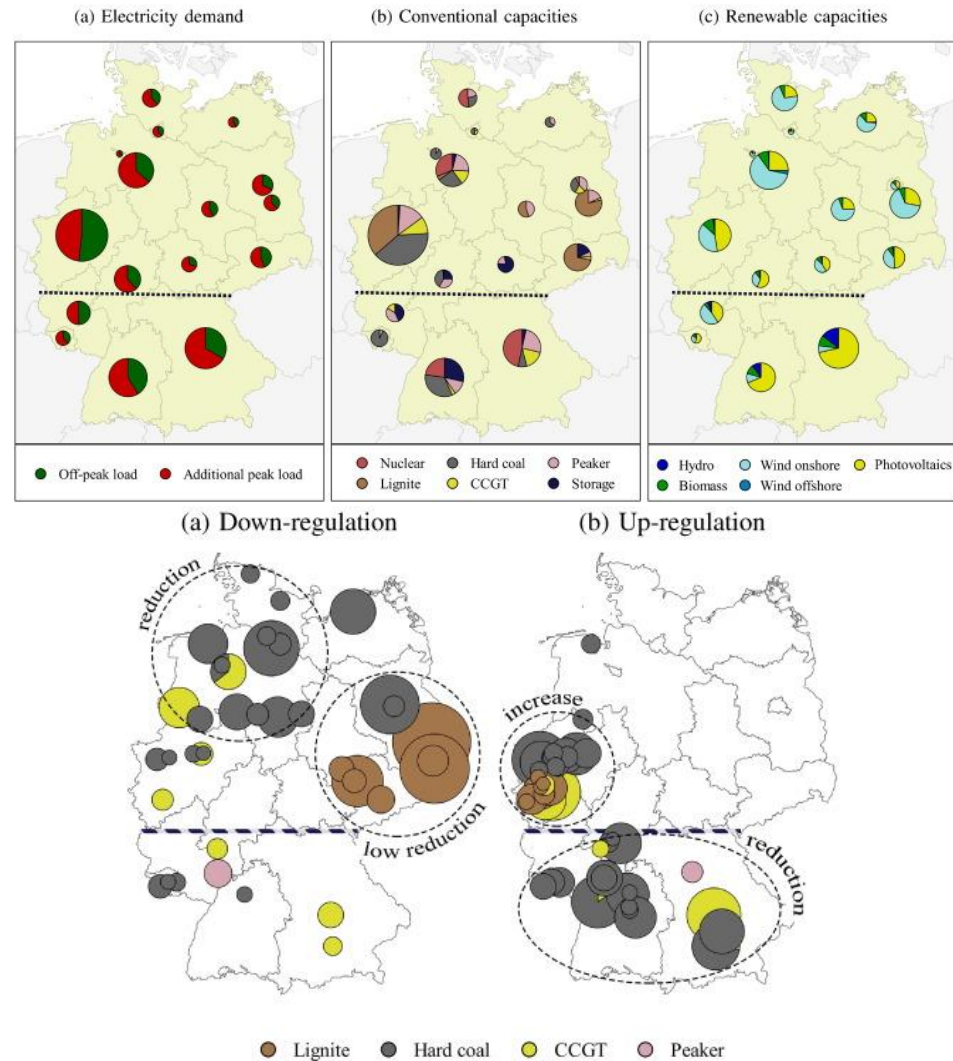
Buy-back
congestion
= cost



Buy-back with
price
discrimination

Germany – counter-trade to mitigate internal imbalance

- Germany today has only 1 price zone
- High production capacity in north, high consumption further down
- In 2023, the TSO payed **4 billion Euros** to mitigate imbalances in counter-trade fashion
- Dividing into several zones would decrease redispatch



Egerer, Jonas, Jens Weibezahn, and Hauke Hermann. "Two price zones for the German electricity market—Market implications and distributional effects." *Energy Economics* 59 (2016): 365-381.

Recap

- Why is congestion management needed in the power system?
- Understand the social benefits of having sufficient interconnection capacity
- Understand the principle of area pricing
- Area pricing as an optimization Problem
- Understand the principles of counter trade or buy-back
- Be able to solve simple cases of area pricing and counter trades in practice
- Understand the advantages and disadvantages of each method